

WorkshopPlus: Azure AI Platform and Services

AI Language Services

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Learning Units covered in this Module



Introduction to AI Language Services



Natural Language Processing: Extract and classify text, understand conversations, AI Translator, Content Understanding/AI Foundry



Choosing the right AI Service



Responsible Use



Containerization, Integration, Real World applications and AI Agents



Lab: AI Language



Introduction to AI Language Services

Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The different Azure AI Language Services
- ✓ How these services work

2

Identify....

- ✓ Use cases for each service



What is Natural Language Processing (NLP)?

Azure AI Language is a cloud-based service that provides **Natural Language Processing (NLP)** features for **understanding** and **analyzing** text.

The screenshot displays the Azure AI Language interface for analyzing a sample email review. The interface includes a sidebar on the left with configuration options, a central text area with the email content and NLP annotations, and a details panel on the right.

Configuration Options (Left Sidebar):

- action**
- version**: 01 (GA)
- el version**: 15-preview
- language**
- s to include**
- licy**
- ngest**
- ptions**
- lize values**
- normalization of detected value in metadata.
- ization provides a standard way ning the value to be consumed er processing, see article: [Learn](#)

Sample Text (Center):

Sample: Email (Review) [Text] [JSON]

Dear Sir/Madam,
P... Per...

My name is Mateo Gomez and I visited Contoso Restaurant with my spouse for our anniversary on Thursday of last week. Prior to my visit, I had heard great things about your food and service from my neighbors, as I actually live right down the street from you at 1234 Hollywood Boulevard. After a lovely stroll along the Santa Monica Pier, we decided to go for dinner at a place we heard so many good things about. When we got to the restaurant last Thursday, we had to wait for 15 minutes just to get on the waiting list. After an hour of waiting to be seated, we ordered two Surf and turf platters. To our absolute surprise, the food was very cold and dry. The whole experience was just quite awful, and I feel like we are owed a refund and an apology for that day. I would appreciate it if you could call me on my number 949-555-0110.

Details (Right Panel):

- Sir**
Type: PersonType
Offset: 5
Length: 3
Confidence: 90%
Tags: PersonType (90%)
- Madam**
Type: PersonType
Offset: 9
Length: 5
Confidence: 81%
Tags: PersonType (81%)
- Mateo Gomez**
Type: Person
Offset: 27
Length: 11
Confidence: 100%
Tags: Person (100%)
- Contoso Restaurant**
Type: Location
Offset: 53

AI language examples

Configuration

Select API version

2024-11-01

Select model

2024-04-15

Select text lan

English

Select types tr

All

Overlap policy

Match long

Inference opti

☒ Normaliz

Include noi
entities val
Normalizat
of returnin
by further j
[more](#)

Configuration

Select API version

2024-11-15

Select model

latest ("GA"

Select text lan

English

Select types t

All

Specify redac

CharacterM

Specify redac

*

Activity log

Access control (IAM)

Tags

Diagnose an

Resource M

Keys and

Encrypti

Pricing t

Network

Identity

Cost and

Properti

Locks

Security

Monitoring

Automation

Help

Configuration

Select API version

2024-11-01 (GA

Select model versi

latest ("GA"

Select text langua

English

Summarization aspects

☒ Recap

Summarizes a conversation into one paragraph

☒ Chapter title

Segments a conversation and generates a title for each segment

☒ Narrative

Segments a conversation and generates a summary for each segment

Sample

Email (Review)

Text

JSON

Details

Sample

Product Troubleshooting Customer Service Call Transcript

Text

JSON

Hide PII

Details

Sample

Product Review

Text

JSON

Details

Sample

Multiple speakers

Text

JSON

Details

["KEVIN SCOTT"]

Thank you so much for doing the show today, and I just wanted to jump right in with maybe one of the more interesting things that's happened in the past few years in technology, which is GPT-4 and ChatGPT and the work that we've been doing together at Microsoft with OpenAI. By the time this podcast airs, OpenAI will have made their announcement to the world about GPT-4, but I want to sort of set the stage. The unveiling of the first instance of GPT-4 outside of OpenAI was actually to you last August at a dinner that you hosted with Reid and Sam Altman and Greg Brockman and Satya and a whole bunch of other folks. The OpenAI folks had been very anxious about showing you this because your bar for AI had been really high, and I think it has been really helpful actually, the push that you had made on all of us for like what acceptable high ambition I would look like. And so, I wanted to ask you, what was that dinner like for you? What were your impressions like? What had you been thinking before, and like what, if anything, changed in your mind after you had seen GPT-4?

["BILL GATES"]

10967 / 12500

Run

Details

General summarization

Recap

- The speaker, a high-profile AI advocate, discussed the advancements in AI technology, specifically GPT-4 and ChatGPT, developed by Microsoft and OpenAI. The speaker highlighted the advancements in AI, particularly in understanding natural language and understanding complex logic. The speaker also highlighted the potential of these systems, particularly in understanding context and problem-solving, and provided an example of how these systems can be used to assist in reasoning. The speaker emphasized the need for further training to address these areas.

Conversational segmentation

Chapter Title

- Progress and Future of Artificial

Microsoft is powered by Azure AI

Applications

Application Platform

AI Builder / Low Code

Scenario-Based Services

Customizable AI Models

ML Platform

 Microsoft 365

 Microsoft Dynamics 365

 Copilot

Partner Solutions



Power BI



Power Apps



Power Automate



Copilot Studio
Copilot Agents



Bot Service



AI Search



Document Intelligence



Video Indexer



Metrics Advisor



Immersive Reader



Vision



Speech



Language



Decision

Azure OpenAI Service
+ Industry Models



Azure Machine Learning



Microsoft Fabric

Microsoft products using Azure AI Language



Bing



Office



SharePoint



Teams



Purview



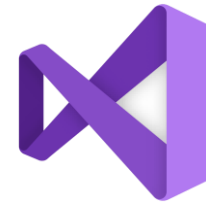
Viva Engage



Dynamics 365



Designer



Visual Studio



LinkedIn

Coming Soon: AI Interpreter Agent

Azure AI Foundry



Copilot Studio



Visual Studio



GitHub



Azure AI
Foundry SDK



Model Catalog

Foundational models

Open-source models

Task models

Industry models



Azure
OpenAI Service



Azure
AI Search



Azure AI
Agent Service



Azure AI
Content Safety



Azure
Machine Learning

Evaluations

Customization

Governance

Monitoring

Observability

AI Foundry

Playgrounds - Azure AI Foundry

https://ai.azure.com/playgrounds

Azure AI Foundry / AI-VBD-Project / Playgrounds

All hubs + projects

Project AI-VBD-Project

Overview

Model catalog

Playgrounds

AI Services

Build and customize

Agents PREVIEW

Templates PREVIEW

Fine-tuning

Prompt flow

Assess and improve

Tracing PREVIEW

Evaluation

Safety + security

My assets

Models + endpoints

Data + indexes

Web apps

Try the Speech playground

Try the Agents playground

Real-time audio playground PREVIEW

Build low-latency conversational experiences with native speech to speech, natural voices, and multimodal input.

Try the Real-time audio playground

Images playground

Generate images from text prompts using AI models like DALL-E.

Try the Images playground

Healthcare playground PREVIEW

Apply multimodal analysis and reasoning on a variety of medical data, including imaging, clinical records, and more.

Try the Healthcare playground

Language playground PREVIEW

Analyze and summarize using LLM-powered natural language processing capabilities.

Try the Language playground

Learning resources

Dive into documentation

Learn about different playgrounds and get all the specs you need to start experimenting.

Try it out with a tutorial

Start with a how-to and watch one, then do one...and you'll be ready to teach one in no time!

Natural Language Processing

NLP Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of natural language processing
- ✓ The principles of text summarisation, translation, information extraction

2

Identify....

- ✓ Use cases for each service



Azure AI Language Services

Extract, Classify, Summarize

- [Extract key phrases](#)
- [Find linked entities](#)
- [Named Entity Recognition \(NER\)](#)
- [Personally Identifiable Information \(PII\) detection](#)
- [Custom Named Entity Recognition \(custom NER\)](#)
- [Text analytics for health](#)
- [Document and Conversation summarization](#)
- [Analyze sentiment and mine text for opinions](#)
- [Detect Language](#)
- [Custom text classification](#)

Translate

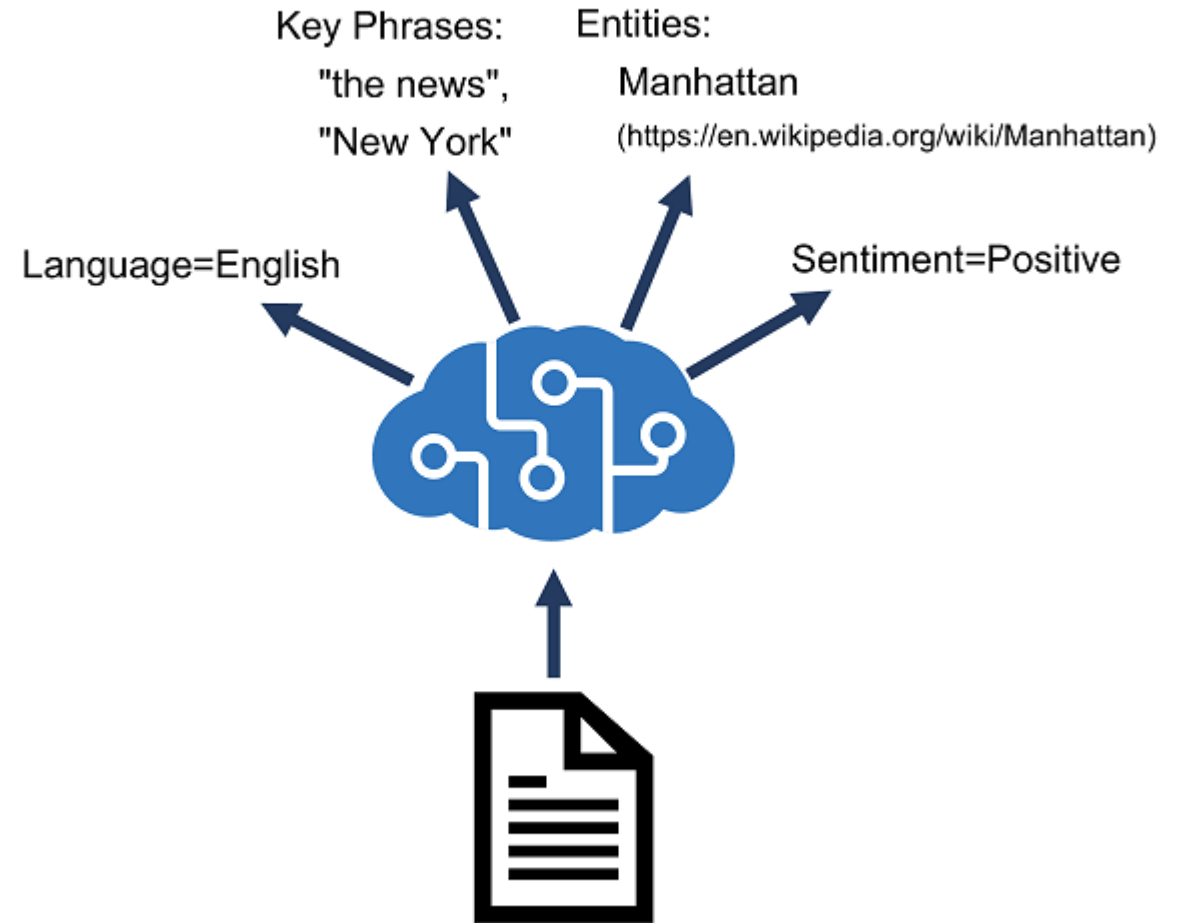
- [Text Translation](#)
- [Document Translation](#)
- [Custom Translator](#)

Language Understanding

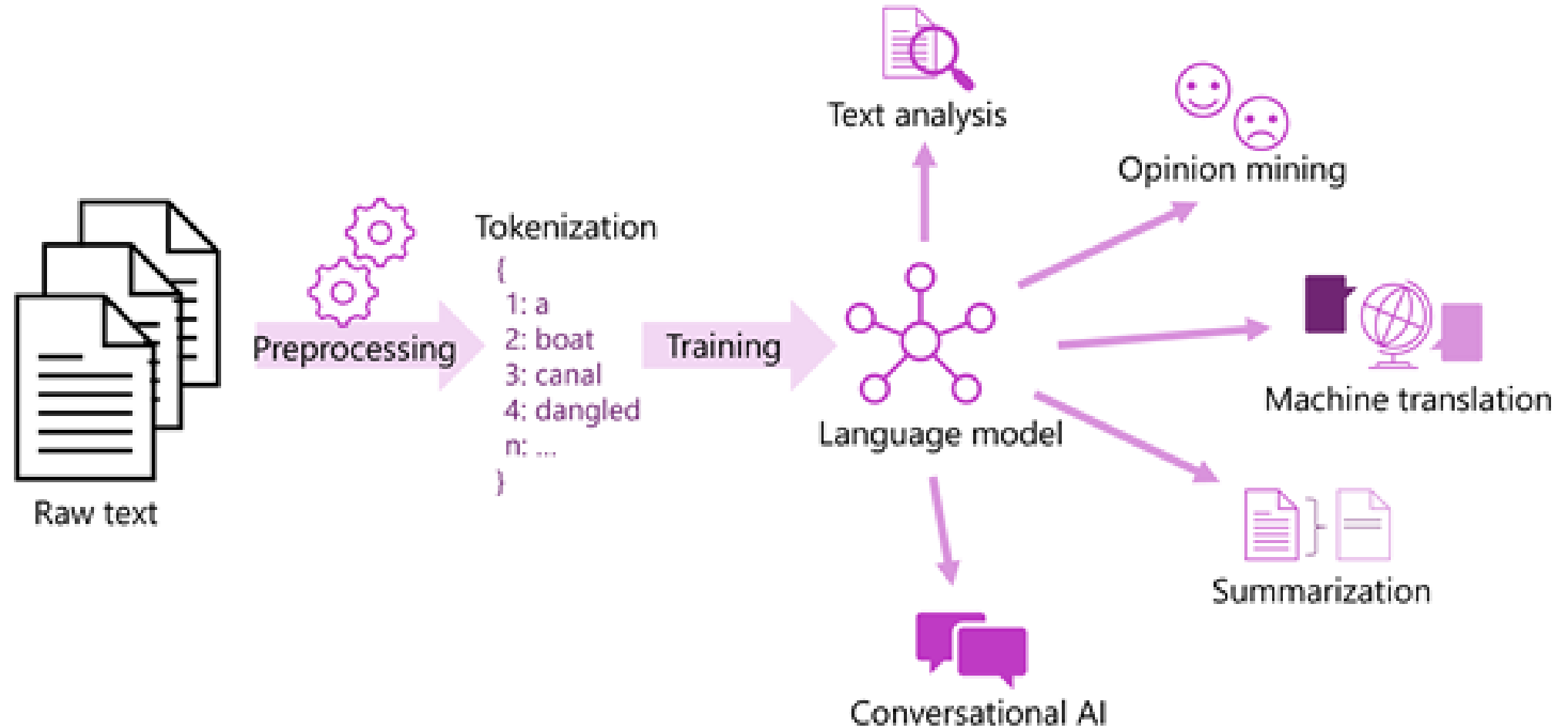
- [Conversational Language Understanding](#)
- [Orchestration Workflow](#)
- [Question Answering](#)

AI Language

Azure AI Language is a cloud-based service that provides Natural Language Processing (NLP) features for understanding and analyzing text.



Natural language processing overview



Tokenization

We choose to go to the moon.

we choose to go to the moon

we choose to go to the moon

we choose ~~to~~ go ~~to~~ ~~the~~ moon

we choose ~~to~~ go ~~to~~ ~~the~~ moon

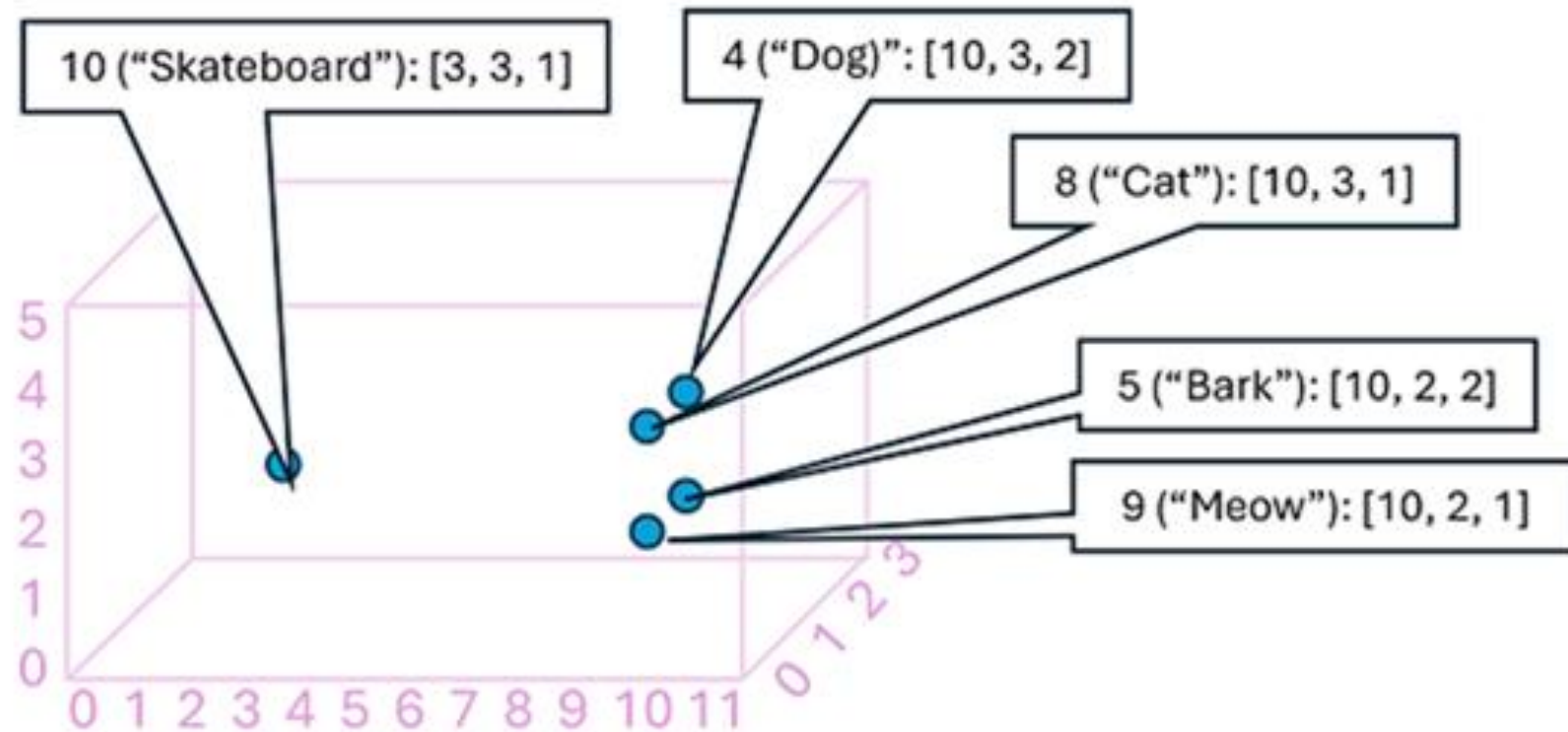
Text normalisation

Tokenization

Stop word removal

n-grams (*bigram*)

Semantic Language Models



Named Entity Recognition (and custom)

- News aggregation
- Analysing customer queries
- Research tool
- Document analysis
- Enhance search capabilities
- Automate business processes
- Customer analysis

The screenshot displays a Named Entity Recognition (NER) interface. At the top, there's a 'Sample' dropdown menu set to 'Legal (NDA)', and two tabs, 'Text' and 'JSON', are visible. The main text area contains a sample document snippet with several entities highlighted in blue and labeled with tags like 'Date', 'Person...', 'Organization', 'Skill', and 'Email'. A tooltip is shown over the 'Date' entity, displaying its value 'August 17th, 2019' and a confidence score of 80.00%. On the right side, a 'Details' panel provides more information about the selected entity, including its type, offset, length, and confidence. Below this, a 'Tags' section shows 'Date (80%)' and 'DateTime (80%)'. A 'Metadata' section lists 'Timex: 2019-08-17' and 'Value: 2019-08-17'. At the bottom of the interface, there's a 'Run' button and a page indicator '665 / 5000'.

Sample: Legal (NDA) Text JSON

On the date of August 17th, 2019, as an employee of Contoso Restaurant, I, Mateo Gomez, residing in CA, with social security number: 123-45-6789, I hereby agree to promote the top priorities delegated to me at Contoso Restaurant, and vow to never disclose any information including but not limited to trade secrets, finances, delivery schedules, and recipes. If I, Mateo Gomez, accidentally or with intent breach the conditions set forth in this contract, understand fully that I shall receive a written termination to the following email address mateo@contosorestaurant.com as

Date
Value: August 17th, 2019
Confidence: 80.00%

Details

August 17th, 2019
Type: Date
Offset: 15
Length: 17
Confidence: 80%

Tags
Date (80%) DateTime (80%)

Metadata
Timex: 2019-08-17
Value: 2019-08-17

employee
Type: PersonType
Offset: 40
Length: 8
Confidence: 99%

Tags
PersonType (99%)

665 / 5000 Run

Note confidence score and ISO formatted date

Key phrase extraction

- Information Retrieval
- Text Summarisation
- Market research
- Customer support categorisation/prioritisation
- Content Recommendation

Key phrases

Pike place market, favorite Seattle attraction

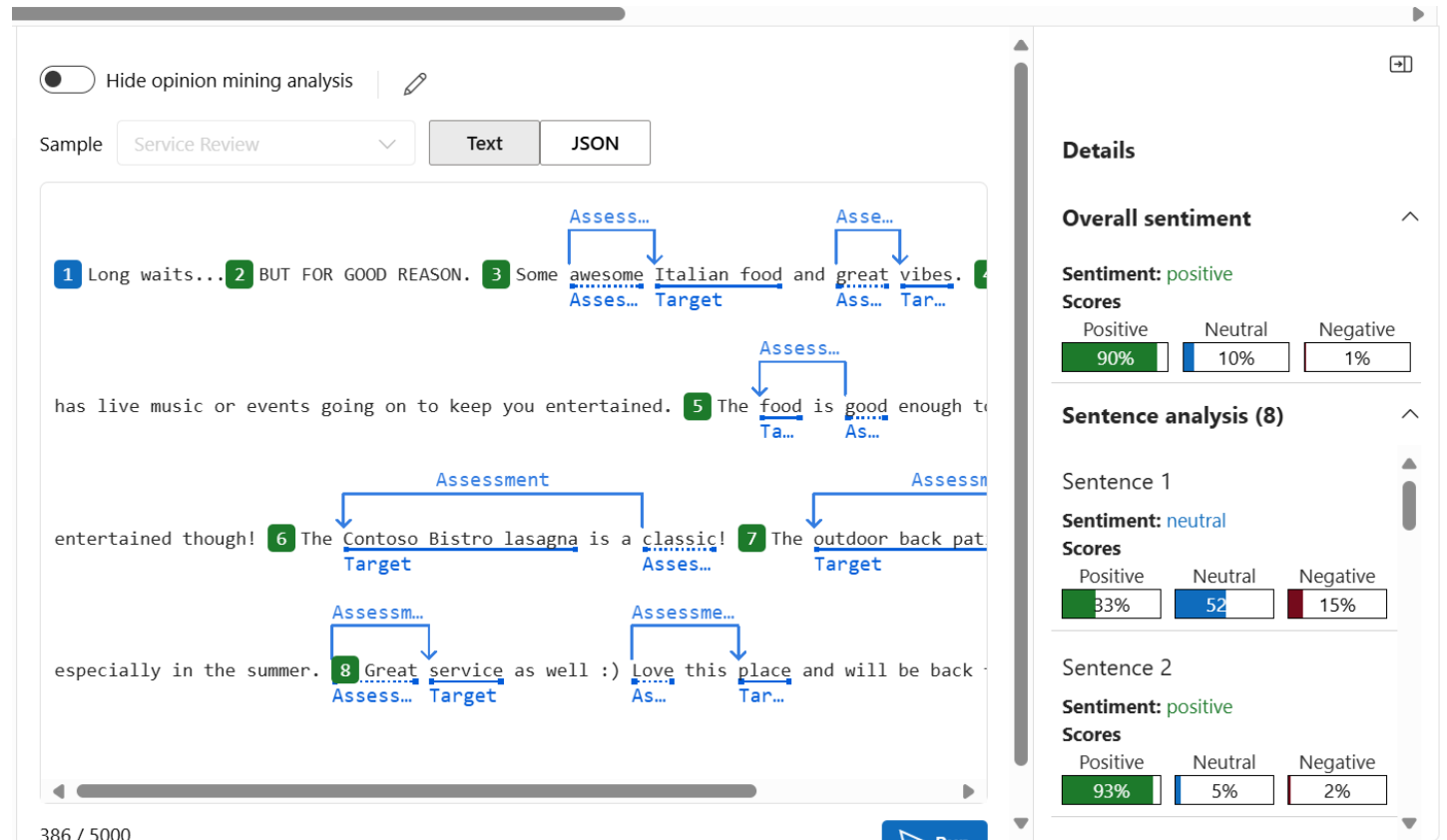
Original text

Pike place market is my favorite Seattle attraction.
Key phrase Key phrase

Sentiment Analysis and Opinion Mining

Example Applications

- Customer feedback analysis
- Social media analysis
- Product development
- Brand monitoring
- Survey analysis
- Customer service improvement
- Marketing campaign analysis



Entity Linking

Example Applications

- Enhanced search engine understanding
- Improved content recommendation
- Information extraction
- Improving performance of chatbots and virtual assistants by linking entities to a knowledge base

The diagram illustrates the process of entity linking. At the top, three separate boxes show individual entities being linked to their respective Wikipedia pages: 'Microsoft' is linked to 'Microsoft -Wikipedia', 'Bill Gates' is linked to 'Bill Gates -Wikipedia', and 'Paul Allen' is linked to 'Paul Allen -Wikipedia'. Below these, the 'Original text' is shown: 'Microsoft was founded by Bill Gates and Paul Allen on April 4, 1975.' Each entity in the sentence is highlighted with a blue underline, and a small tooltip appears below each underline, displaying the text 'Linked entity'.

Linked entity
[Microsoft -Wikipedia](#)

Linked entity
[Bill Gates -Wikipedia](#)

Linked entity
[Paul Allen -Wikipedia](#)

Original text

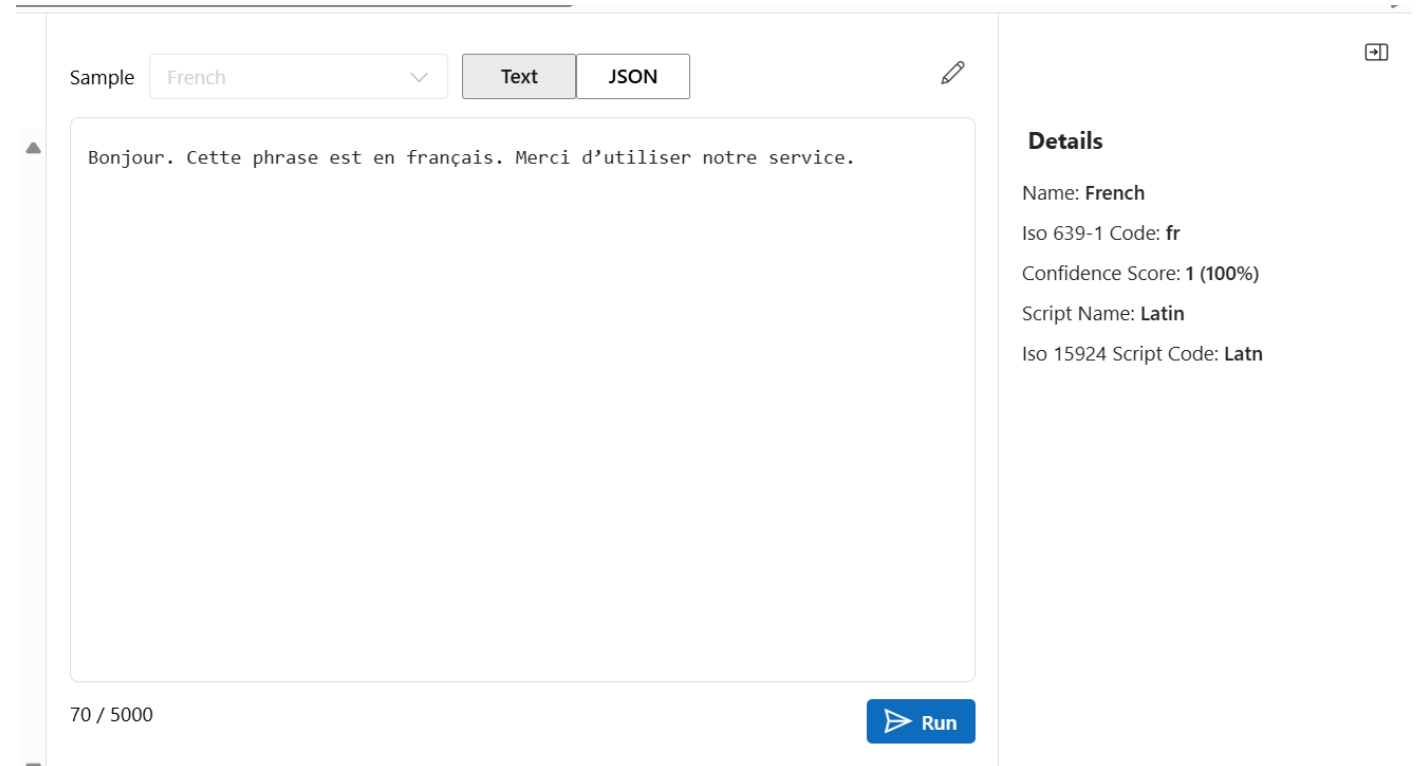
[Microsoft](#) was founded by [Bill Gates](#) and [Paul Allen](#) on [April 4](#), 1975.

Linked enti... Linked entity Linked entity Linked e...

Language Detection

Example Applications

- Customer service routing
- Content translation
- Security and compliance (eg for PII)
- Social media analysis
- Identify language of reviews
- Document management



The screenshot shows a web application for language detection. At the top, there is a 'Sample' dropdown menu set to 'French', followed by two buttons: 'Text' (selected) and 'JSON'. To the right of these buttons is a small edit icon. Below this is a large text input area containing the French sentence: 'Bonjour. Cette phrase est en français. Merci d'utiliser notre service.' At the bottom left of the input area, it says '70 / 5000'. At the bottom right, there is a blue 'Run' button with a play icon. On the right side of the interface, there is a 'Details' section with the following information: Name: French, Iso 639-1 Code: fr, Confidence Score: 1 (100%), Script Name: Latin, and Iso 15924 Script Code: Latn.

Sample: French [v] Text JSON [edit icon]

Bonjour. Cette phrase est en français. Merci d'utiliser notre service.

70 / 5000 [Run]

Details

Name: French
Iso 639-1 Code: fr
Confidence Score: 1 (100%)
Script Name: Latin
Iso 15924 Script Code: Latn

SDK and REST API for the Language Service

Client libraries available for

- .NET
- Java
- Javascript
- Python

<https://learn.microsoft.com/en-us/azure/ai-services/language-service/concepts/developer-guide?tabs=language-studio>

Demonstration

AI Language Extraction

The screenshot displays the Azure AI Foundry interface for the 'AI-Language-Project' under the 'Language Playground' section. The top navigation bar includes 'All hubs + projects', a search icon, and a user profile icon. The left sidebar contains a navigation menu with the following items: Overview, Model catalog, Playgrounds (selected), AI Services, Build and customize, Agents (PREVIEW), Templates (PREVIEW), Fine-tuning, Prompt flow, Assess and improve, Tracing (PREVIEW), Evaluation, Safety + security, My assets, Models + endpoints, Data + indexes, Web apps, and Management center.

The main content area features a horizontal scrollable list of AI language extraction tasks:

- Analyze sentiment**: Detect positive, negative and neutral sentiment in text. Get more insights by mining opinions.
- Detect language**: Evaluate text and detect a wide range of languages and variant dialects.
- Extract health information**: Extract and label relevant health information from unstructured text.
- Extract key phrases**: Identify the most important points in a piece of text.
- Extract named entities**: Identify different entities in text and categorize them into pre-defined types.
- Extract PII from**: Identify sensitive information associated with types.

Below the task list, the 'Configuration' section is visible, showing the following settings:

- Select API version**: 2024-11-15-preview
- Select model version**: 2024-04-15-preview
- Select text language**: English
- Select types to include**: All
- Specify redaction policy**: CharacterMask
- Specify redaction character**: (empty field)

The 'Sample' section includes a dropdown menu 'Select a sample...' and a large text area with the placeholder text 'Select a sample, upload a file or enter your text here'. The text area shows a character count of '0 / 5000'. A 'Run' button is located at the bottom right of the text area.

The 'Details' section on the right contains the text: 'Your details will appear after you enter or upload some text and press Run.'

Summarization

Summarization Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of summarization
- ✓ The principles of extractive and abstractive summarization

2

Identify....

- ✓ Use cases for each service



Summarize

Example Applications

- Customer service summaries
- Meeting summaries
- Content creation
- Document management
- News aggregation
- Summarise educational content

Configuration

Select API version

2024-11-01 (GA) ▾

Select text language

English ▾

Summarization aspects

☒ Recap

Summarizes a conversation into one paragraph

☒ Chapter title

Segments a conversation and generates a title for each segment

☒ Narrative

Segments a conversation and generates a summary for each segment

Sample

Multiple speakers ▾

Text

JSON

["KEVIN SCOTT"]

Thank you so much for doing the show today, and I just wanted to jump right in more interesting things that's happened in the past few years in technology, ChatGPT and the work that we've been doing together at Microsoft with OpenAI. E podcast airs, OpenAI will have made their announcement to the world about GPT- of set the stage. The unveiling of the first instance of GPT-4 outside of Open last August at a dinner that you hosted with Reid and Sam Altman and Greg Br whole bunch of other folks. The OpenAI folks had been very anxious about showin your bar for AI had been really high, and I think it has been really helpful a

10967 / 12500

Run

Extractive and Abstractive Summarization

**Abstractive text
summarization now
using Phi-3.5-mini!**

Input

"At Microsoft, we are on a quest to advance AI beyond existing techniques, by taking a more holistic, human-centric approach to learning and understanding. As Chief Technology Officer of Azure AI services, I have been working with a team of amazing scientists and engineers to turn this quest into a reality. In my role, I enjoy a unique perspective in viewing the relationship among three attributes of human cognition: monolingual text (X), audio or visual sensory signals, (Y) and multilingual (Z). At the intersection of all three, there's magic—what we call XYZ-code as illustrated in Figure 1—a joint representation to create more powerful AI that can speak, hear, see, and understand humans better. We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages. The goal is to have pretrained models that can jointly learn representations to support a broad range of downstream AI tasks, much in the way humans do today. Over the past five years, we achieve human performance on benchmarks in conversational speech recognition, machine translation, conversational question answering, machine reading comprehension, and image captioning. These five breakthroughs provided us with strong signals toward our more ambitious aspiration to produce a leap in AI capabilities, achieving multi-sensory and multilingual learning that is closer in line with how humans learn and understand. I believe the joint XYZ-code is a foundational component of this aspiration, if grounded with external knowledge sources in the downstream AI tasks."

Extractive

- "At Microsoft, we are on a quest to advance AI beyond existing techniques, by taking a more holistic, human-centric approach to learning and understanding."
- "We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages."
- "The goal is to have pretrained models that can jointly learn representations to support a broad range of downstream AI tasks, much in the way humans do today."

Abstractive

"Microsoft is taking a more holistic, human-centric approach to learning and understanding. We believe XYZ-code enables us to fulfill our long-term vision: cross-domain transfer learning, spanning modalities and languages. Over the past five years, we achieved human performance on benchmarks in conversational speech recognition."

Conversation Summarization

Input

Agent: "Hello, you're chatting with Rene. How may I help you?"

Customer: "Hi, I tried to set up wifi connection for Smart Brew 300 espresso machine, but it didn't work."

Agent: "I'm sorry to hear that. Let's see what we can do to fix this issue. Could you push the wifi connection button, hold for 3 seconds, then let me know if the power light is slowly blinking?"

Customer: "Yes, I pushed the wifi connection button, and now the power light is slowly blinking."

Agent: "Great. Thank you! Now, please check in your Contoso Coffee app. Does it prompt to ask you to connect with the machine?"

Customer: "No. Nothing happened."

Agent: "I see. Thanks. Let's try if a factory reset can solve the issue. Could you please press and hold the center button for 5 seconds to start the factory reset."

Customer: "I've tried the factory reset and followed the above steps again, but it still didn't work."

- Agent:** "I'm very sorry to hear that. Let me see if there's another way to fix the issue. Please hold on for a minute."

Output

Example summary	Remark	Conversation aspect
Customer is unable to set up wifi connection for Smart Brew 300 espresso machine	a customer issue in a customer-and-agent conversation	issue
The agent suggested several troubleshooting steps, including checking the wifi connection, checking the Contoso Coffee app, and performing a factory reset. However, none of these steps resolved the issue. The agent then put the customer on hold to look for another solution.	solutions tried in a customer-and-agent conversation	resolution
The customer contacted the agent for assistance with setting up a wifi connection for their Smart Brew 300 espresso machine. The agent guided the customer through several troubleshooting steps, including a wifi connection check, checking the power light, and a factory reset. Despite following these steps, the issue persisted. The agent then decided to explore other potential solutions	Summarizes a conversation into one paragraph	recap
Troubleshooting SmartBrew 300 Espresso Machine	Segments a conversation and generates a title for each segment; usually cowork with narrative aspect	chapterTitle
The customer is having trouble setting up a wifi connection for their Smart Brew 300 espresso machine. The agent suggests several solutions, including a factory reset, but the issue persists.	Segments a conversation and generates a summary for each segment, usually cowork with chapterTitle aspect	narrative

Document Summarization - *Preview*

File type	File extension	Description
Text	.txt	An unformatted text document.
Adobe PDF	.pdf	A portable document file formatted document.
Microsoft Word	.docx	A Microsoft Word document file.

Why does Azure AI Language's Summarization Service matter?

Premium Quality at Affordable Price: The combination of Generative Language AI models (Azure Open AI service) and task-optimized encoder models offers text summarization solutions with higher quality, cheaper costs, and lower latency.

1. Azure AI Language's Summarization - Key Customer Scenarios

1.1 Summarization - Conversation



Contact Center Call
& Meeting Analytics



1.2 Summarization - Document



Document Search
& Document Discovery



1. Azure AI Language's Summarization - Key Highlights



10x Cost Savings: Azure AI Language Summarization in English costs ~0.008/1k tokens vs. \$0.06/1k tokens for AOAI/GPT4-32k



Leading Quality with fine-tuned LLM for both document and conversation summarization with built-in hallucination detection.



More Capabilities: Extractive (encoder model) + **Abstractive** summarization enables more use cases than decoder-only Gen AI



No Prompt Engineering needed because of out-of-the-box summarization styles and aspects with native document support



Disconnected & Connected Container offerings with both GPU & CPU options with distilled models for optimal compute usage

Summarization Demonstration

Summarization in AI Language

Summarization in Azure AI Language

- ✓ *Optimized for summarization tasks*
- ✓ *Ready for enterprise deployment*
- ✓ *Powered by GPT*

PII Service

PII Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of Personal Identifying Information (PII) detection
- ✓ The principles of PII Redaction/Removal

2

Identify....

- ✓ Use cases for each service



PII Detection

URL

Entity value:
www.contososteakhouse.com
Confidence: 80.00%

PhoneNumber

Entity value: 312-555-0176
Confidence: 80.00%

Email

Entity value:
order@contososteakhouse.com
Confidence: 80.00%

Organization

Entity value:
contososteakhouse
Confidence: 46.00%

Original text

You can even pre-order from their online menu at www.contososteakhouse.com, call 312-555-0176 or

URL

PhoneNumber

send email to order@contososteakhouse.com!

Organization

Email

View code

View documentation

Connected to ai-admin1650ai362209100...

All

Extract Information

Summarize Information

Classify Text

Named entities

Identify different entities in text and group them into pre-defined categories

Extract PII from conversation

Identify sensitive entities in text that are associated with an individual.

Extract PII from text

Identify sensitive entities in text that are associated with an individual.

Summarize conversation

Recap conversation and segment long meetings into timestamps chapters

Summarize for call center

Recap call and summarize for customer issues and resolution

Summarize text

Summarize and extract key information at scale from text

Configuration

Select API version

2024-11-15-preview

Select model version

2024-04-15-preview

Select text language

English

Select types to include

All

Specify redaction policy

CharacterMask

Specify redaction character

*

Sample

Legal (NDA)

Text

JSON

Hide PII

On the date of August 17th, 2019, as an employee of Contoso Restaurant, I, Mateo Gomez, residing in 1234 Hollywood Boulevard Los Angeles CA, with social security number: 123-45-6788, hereby declare to fully support and promote the top priorities delegated to me at Contoso Restaurant, and vow to never disclose any information including but not limited to trade secrets, finances, delivery schedules, and recipes.

If I, Mateo Gomez, accidentally or with intent breach the conditions set forth in this contract, understand fully that I shall receive a written termination to the following email address mateo@contosorestaurant.com as well as a fine of up to

665 / 5000

Details

August 17th, 2019

Type: Date

Offset: 15

Length: 17

Confidence: 99%

Tags

Date (99%)

DateTime (99%)

employee

Type: PersonType

Offset: 40

Length: 8

Confidence: 99%

Tags

PersonType (99%)

Contoso Restaurant

Type: Organization

Run

Data Anonymization

Challenge

Large volume of documents, forms and user queries with sensitive info poses high risks of data leaks.

Solution

Build an automated process to extract and anonymize sensitive information in documents, forms or user queries. Manage the extracted sensitive info in a separate, more secured environment.

Azure AI Services

- Personal Identifiable Information (PII) Detection with built-in native doc processing support
- Document Intelligence service

Impact

Data is protected more securely and efficiently.



Personally Identifiable Information (PII) Detection

- Identifies, categorizes, and redacts sensitive info in unstructured text
- Supports a wide range of PII info, including region-specific PII, PHI and PCI
- Supports conversational scenario with a specialized model (aka. Conversation PII)
- 79 languages supported

Typical use cases:

- Mask sensitive info to avoid the leak, inappropriate handling or data bias
- Auto label and classify docs based on the sensitivity

Sample text: The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com , call 312-555-0176 or send email to order@contosohouse.com !	Recognized entity categories: URL PhoneNumber Email
--	---

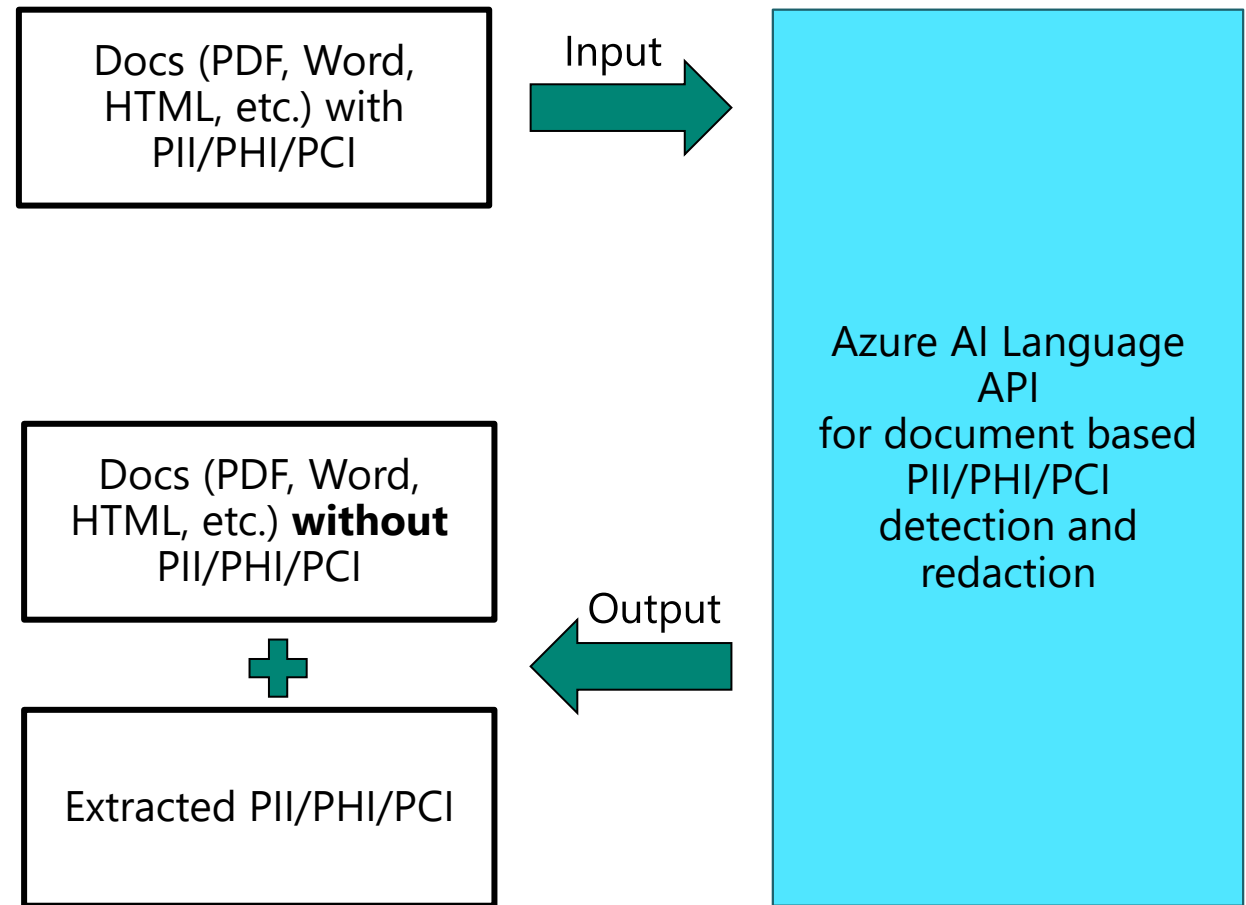
Redacted text: <i>The food is great, and the place was impeccably clean. You can order from their online menu at www.contosohouse.com, call ***** ***** or send email to *****!",</i>

PII Detection – *Supported Categories*

Basic PII Info	Government & Country/Region-Specific	Azure Information
Person	U.S. Social Security Number (SSN)	Azure DocumentDB Auth Key
Person Type	U.S. Driver's License Number	Azure IAAS DB Connection String
Phone Number	U.S. Passport Number	Azure SQL Connection String
Organization – <i>incl. types:</i>	U.S. Individual Taxpayer Identification Number (ITIN)	Azure IoT Connection String
- Medical - Stock Exchange - Sports	U.S. Drug Enforcement Agency (DEA) Number	Azure Publish Setting Password
Address	U.S. Bank Account Number	Azure Redis Cache Connection String
Email	+ <i>more than 90 PII entities across 31 other countries/regions</i>	Azure SAS
URL		Azure Service Bus Connection String
IP Address	Financial Account	Azure Storage Account Key
Date Time – <i>incl. type:</i>	ABA Routing Number	SQL Server Connection String
Date	SWIFT Code	
Quantity – <i>incl. type:</i>	Credit Card Number	
Age	IBAN Code	

PII Redaction on Native Document Formats (preview)

- Identifies and redacts sensitive PII info directly in native documents –
No more doc pre-processing and post-processing needed!
- PII entities and values extracted from documents are included in the API response payload, which can be stored separated and used for later



Why does Azure AI Language's PII Redaction matter?

Text-based PII unblocks adoption of Gen AI use cases by addressing the increased demand for PII. In fact, OpenAI's ChatGPT leverages Azure AI Language's PII service to preserve data privacy.

2. Azure AI Language' PII Service – Key Customer Scenarios

PII Redaction



Data Anonymization



2. Azure AI Language's PII Redaction – Key Highlights

2.1 Text-based PII Redaction

2.2 Native Document-based PII Redaction

2.2 Speech-based PII Redaction

TM

Differentiates Azure's Generative AI offerings from those of competitors (Google, AWS) by offering one solution for **Text-based PII, Document-based PII, and Conversational PII**



Document-based PII reduces the complexity to implement a PII detection and redaction solution by eliminating the need for document pre- or post-processing with **Single API call**



Conversational PII optimized for speech transcription use cases offers leading redaction quality to ensure sensitive information is redacted from both audio and speech transcript.



Ensures privacy and regulatory compliance for customers in industries such as insurance, finance, healthcare, and public sector to identify, detect & redact PII in large volume of data

PII Demonstration

PII Detection and Removal

The screenshot displays the Azure AI Foundry interface for the 'AI-Language-Project' under the 'Playgrounds' section, specifically the 'Language Playground'.

Navigation and Breadcrumbs:

- Azure AI Foundry / AI-Language-Project / Playgrounds / Language Playground
- Project: ai-language-project

Left Sidebar (Navigation):

- Overview
- Model catalog
- Playgrounds**
- AI Services
- Build and customize
 - Agents (PREVIEW)
 - Templates (PREVIEW)
 - Fine-tuning
 - Prompt flow
- Assess and improve
 - Tracing (PREVIEW)
 - Evaluation
 - Safety + security
- My assets
 - Models + endpoints
 - Data + indexes
 - Web apps
- Management center

Main Content Area:

Filter Tabs: All, Extract Information, Summarize Information, Classify Text

Available Actions:

- Analyze sentiment:** Detect positive, negative and neutral sentiment in text. Get more insights by mining opinions.
- Detect language:** Evaluate text and detect a wide range of languages and variant dialects.
- Extract health information:** Extract and label relevant health information from unstructured text.
- Extract key phrases:** Identify the most important points in a piece of text.
- Extract named entities:** Identify different entities in text and categorize them into pre-defined types.
- Extract PII from:** Identify sensitive information associated with text.

Configuration Panel:

- Configuration:**
 - Select API version: 2024-11-15-preview
 - Select model version: 2024-04-15-preview
 - Select text language: English
 - Select types to include: All
 - Specify redaction policy: CharacterMask
 - Specify redaction character: (empty)
- Sample:** Select a sample... (dropdown)
- Text Input:** Select a sample, upload a file or enter your text here (0 / 5000 characters)
- Run:** Run button
- Details:** Your details will appear after you enter or upload some text and press Run.

Text Analytics for Health

Text Analytics for Health Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of the Text Analytics for Health entity recognition service
- ✓ The principles of relation and assertion in Text Analytics for Health

2

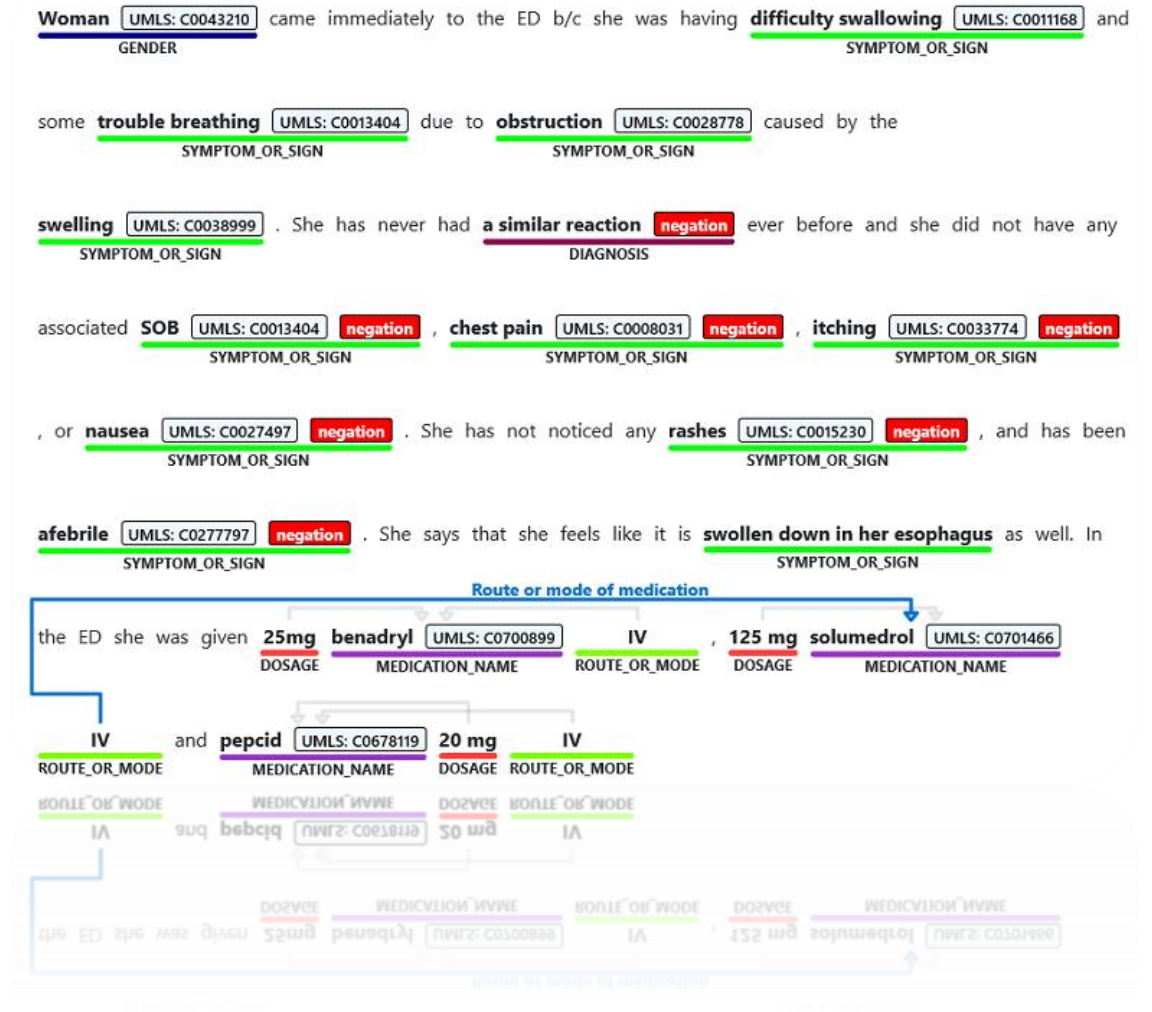
Identify....

- ✓ Use cases for this service



Available in containers

- **Entity recognition**
(30+ healthcare related entity types)
- **Entity linking**
(to UMLS Metathesaurus Knowledge Source)
- **Relation extraction** (35 relation types)
- **Assertion detection**
(Certainty, Conditional, Association, Temporal)
- 7 languages supported (6 languages in preview)



Demo: Text Analytics for Health

1 Named Entity Recognition:

Key terms highlighted & labelled

| 054351 | 2/14/2001 12:00:00 AM | CORONARY ARTERY DISEASE
Date Time Diagnosis

Signed Discharge Date: 4/24/2001 ADMISSION DIAGNOSIS:
AdministrativeEvent Adminis...

2 Entity Linking: Terms linked to respective UMLS codes

open heart surgery
TreatmentName

UMLS: C01897...
✓ UMLS: C0189745
CHV: 0000019333
MDR: 10048935
MEDLINEPLUS: 4119
SNMI: P1-31004

3 Relation Extraction:

Related terms associated together

2:30 am this morning of a sore throat UMLS: C0242429
TIME SYMPTOM_OR_SIGN

Time of condition

4 Assertion Detection:

Terms also labelled with contextual modifiers
(i.e. certainty, conditionality, association)

but remains unsure if she wants to start adjuvant hormonal therapy Hypothetical . Please hold
TREATMENT_NAME

lactulose UMLS: C0022957 Negative Conditional if diarrhea UMLS: C0011991 Hypothetical worsen
MEDICATION_NAME SYMPTOM_OR_SIGN CONDITION_QUALIFIER

Text Analytics for Health Demonstration

Text Analytics for Health

The screenshot displays the Azure AI Foundry interface for the 'AI-Language-Project' under the 'Playgrounds' section, specifically the 'Language Playground'. The interface is divided into several sections:

- Navigation Sidebar:** Includes links for Overview, Model catalog, Playgrounds (selected), AI Services, Build and customize, Agents (PREVIEW), Templates (PREVIEW), Fine-tuning, Prompt flow, Assess and improve, Tracing (PREVIEW), Evaluation, Safety + security, My assets, Models + endpoints, Data + indexes, Web apps, and Management center.
- Top Tabs:** All, Extract Information, Summarize Information, and Classify Text. The 'Extract Information' tab is active.
- Analytics Cards:** A row of six cards representing different text analysis capabilities:
 - Analyze sentiment:** Detect positive, negative and neutral sentiment in text. Get more insights by mining opinions.
 - Detect language:** Evaluate text and detect a wide range of languages and variant dialects.
 - Extract health information:** Extract and label relevant health information from unstructured text.
 - Extract key phrases:** Identify the most important points in a piece of text.
 - Extract named entities:** Identify different entities in text and categorize them into pre-defined types.
 - Extract PII from:** Identify sensitive information associated with types.
- Configuration Panel:** Located below the cards, it includes settings for:
 - Select API version:** 2024-11-15-preview
 - Select model version:** 2024-04-15-preview
 - Select text language:** English
 - Select types to include:** All
 - Specify redaction policy:** CharacterMask
 - Specify redaction character:** (field is empty)
- Sample Input Area:** A large text box with a placeholder 'Select a sample, upload a file or enter your text here'. Below the box is a character count '0 / 5000' and a 'Run' button.
- Details Panel:** On the right, it states 'Your details will appear after you enter or upload some text and press Run.'

Language Understanding

Language Understanding Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of the conversational language understanding service
- ✓ The principles of training data in the conversational language understanding service

2

Identify....

- ✓ Use cases for this service



Conversational Language Understanding (CLU)

Example Applications

- End to end conversational bot
- Human assistant bots
- Command and control application
- Enterprise chat bot

Select text language ⓘ *

English (US) ▼

Deployment name

EmailModel ▼

Enter your own text, or upload a text document

Delete an email

Result

JSON

Intent

Top intent

Delete

Confidence: 96.20%

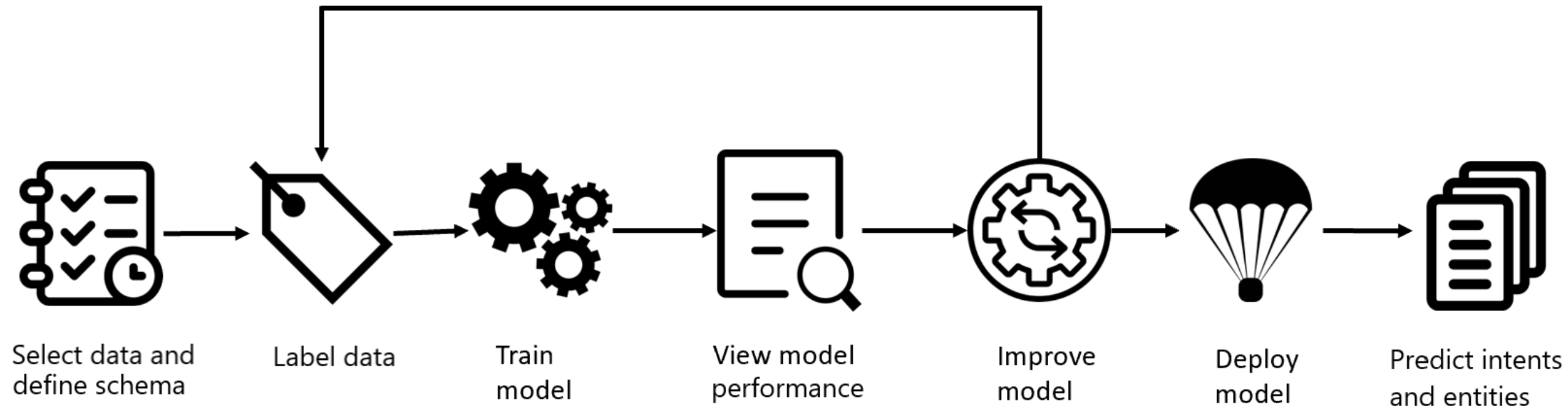
Entities

No entities predicted

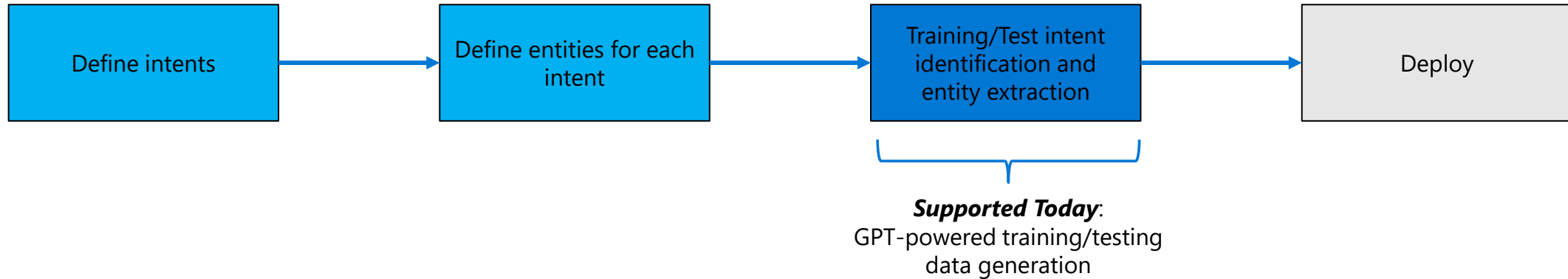
Original text

Delete an email

Conversational Language Understanding



CLU model training in Language Studio with AOAI



Cognitive Services | Language Studio

Language Studio > Conversational Language Understanding projects > Flight_Booking_Demo - Data labeling

Data labeling ✓ Saved

To add an utterance, first select an intent that your user intends to accomplish. Then provide an example of what they'd say in that situation and label the information to extract as entities. After labeling the utterances, you'll be ready to create a model with this data in [Training](#).
For example, for an intent of Order Pizza, an utterance might be "I'd like to get a large pizza." The word "large" might then be labeled as a Pizza Size entity.

Training set **Testing set**

Save changes Upload utterance file Suggest utterances Move to Edit Delete Accept all Reject all Filter

We'll use these utterances to create your conversation model during training. A separate set of utterances can test the performance of your model. [Learn more about splitting data between training and testing sets.](#)

Intent	Utterance
Select Intent	Write your example utterance and press enter
Request Refund	I want my money back!
Request Refund	Get me a refund for that Tuesday flight please
Request Refund	Why have I not gotten any money back for a flight that was changed last minute?
Request Refund	You should issue me a refund as soon as possible
Request Refund	I'd like a refund for my flight that just got cancelled

Suggest utterances with GPT

You can use GPT to suggest utterances for an intent. After the utterances are generated, you can review or edit the suggestions.

Select Azure OpenAI resource

In order to use GPT to suggest utterances, you must select an Azure OpenAI resource and a current deployment from that resource.

Select your Azure Open AI resource*
AOAILeithyTest

You have a connected Azure Open AI resource

Select your Azure Open AI deployment*
textdavinci003

We recommend selecting a deployment with the model "text-davinci-002" for the best results.

Pick your intent

Select your intent and the system will use Azure OpenAI to generate a few utterances for that intent

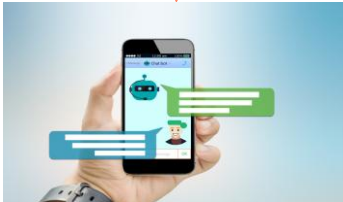
Select intent for suggestions*
Book Flight

Generate utterances Cancel

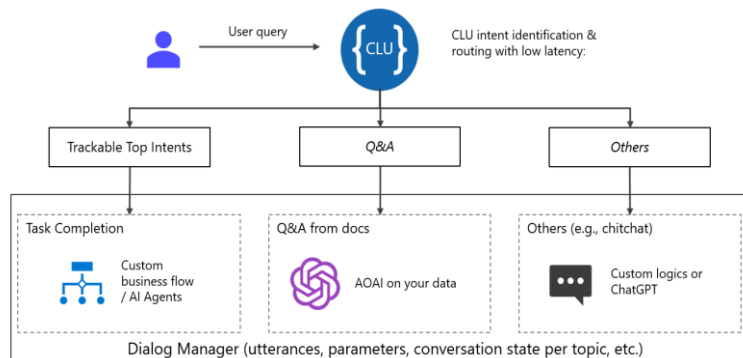
Why does Azure AI Language's CLU + AOAI Orchestration Service matter?

3. Azure AI Language's CLU & AOAI Orchestration – Key Customer Scenarios

CLU & AOAI Orchestration



Chat Intelligence



3. CLU & AOAI Orchestration – Key Highlights



Enhance Chat Intelligence by orchestrating CLU to track intent deflection rate with CLU's in-built **NER** model to improve chat experience with higher quality, cheaper cost and lower latency.



Trigger AI Agents to execute multiple custom business flows and complete tasks based on trackable **top intents** and user-defined **routing logic** to complete end-to-end **business process** tasks.



Orchestrating CLU with Azure OpenAI Service enables encoder model to extract entities from user utterances with higher accuracy



Intent classification in CLU ensures consistent intent schema while using LLMs to identify any new intents from long-tail topics

AI Translator

AI Translator Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of the AI Translator service
- ✓ The principles of information flow within the AI Translator service

2

Identify....

- ✓ Use cases for this service



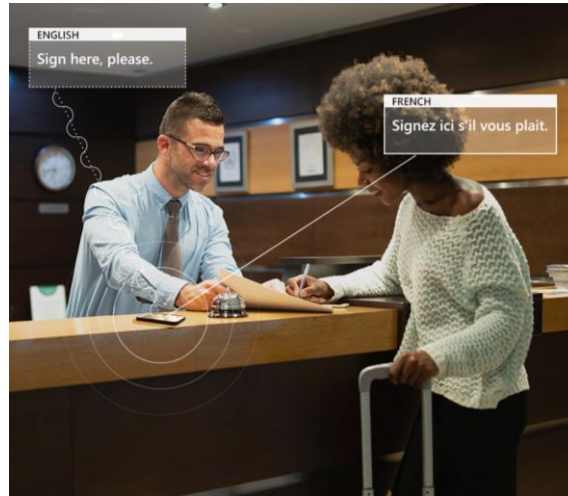
Business Challenges

User-Generated Content / Social Media



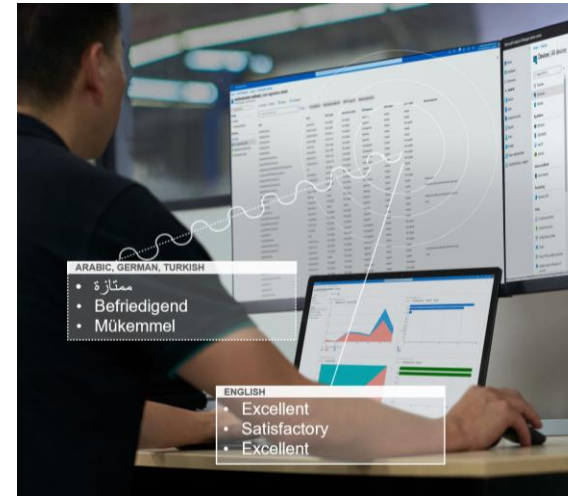
How to I reach all my customers in their preferred language

Localized Collateral



Are my documents localized making it easy to do business

E-Discovery



Content for business intelligence is in different languages across the web

Training Global Teams

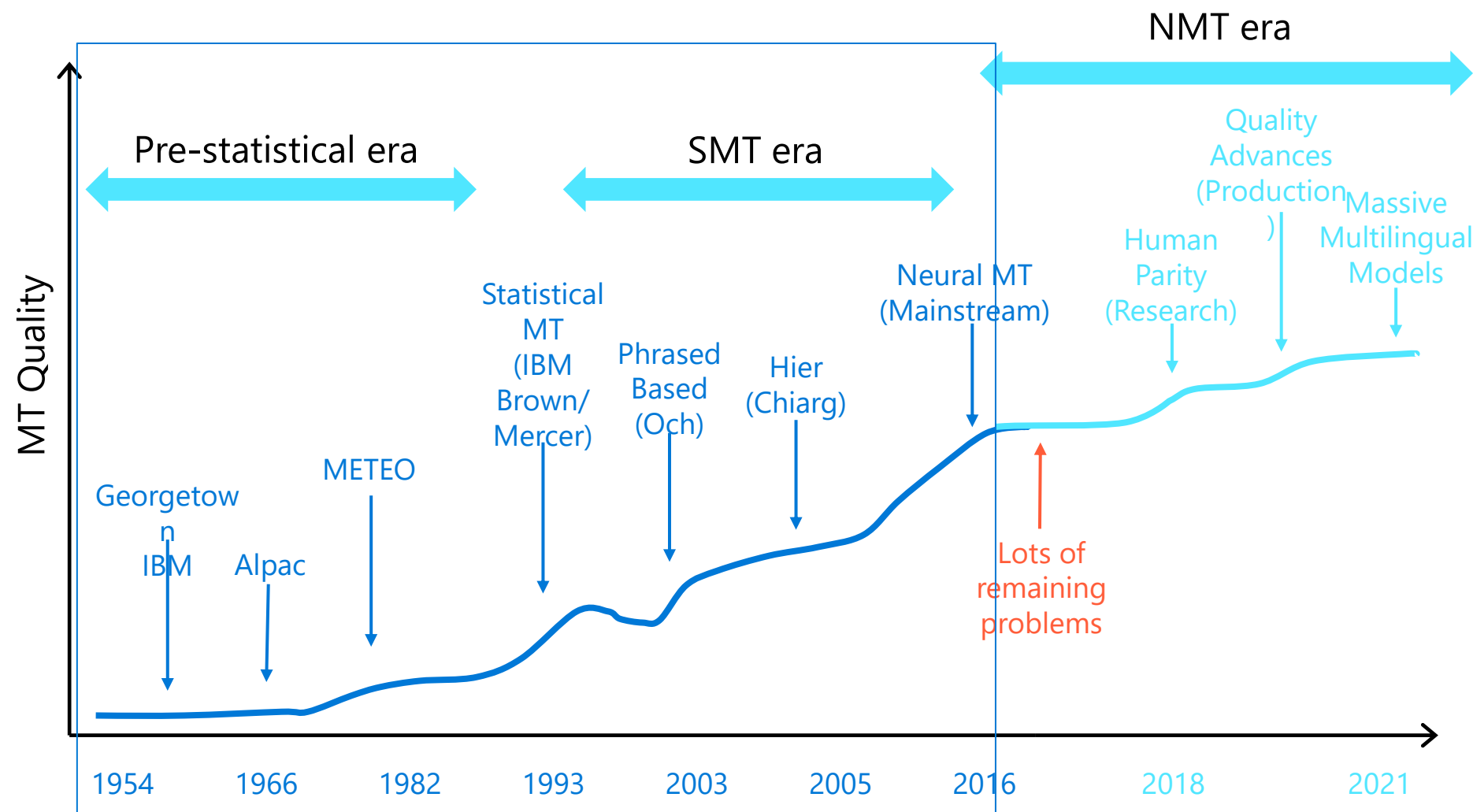


Not everyone has the same level of fluency in all languages

Translator Languages (133)

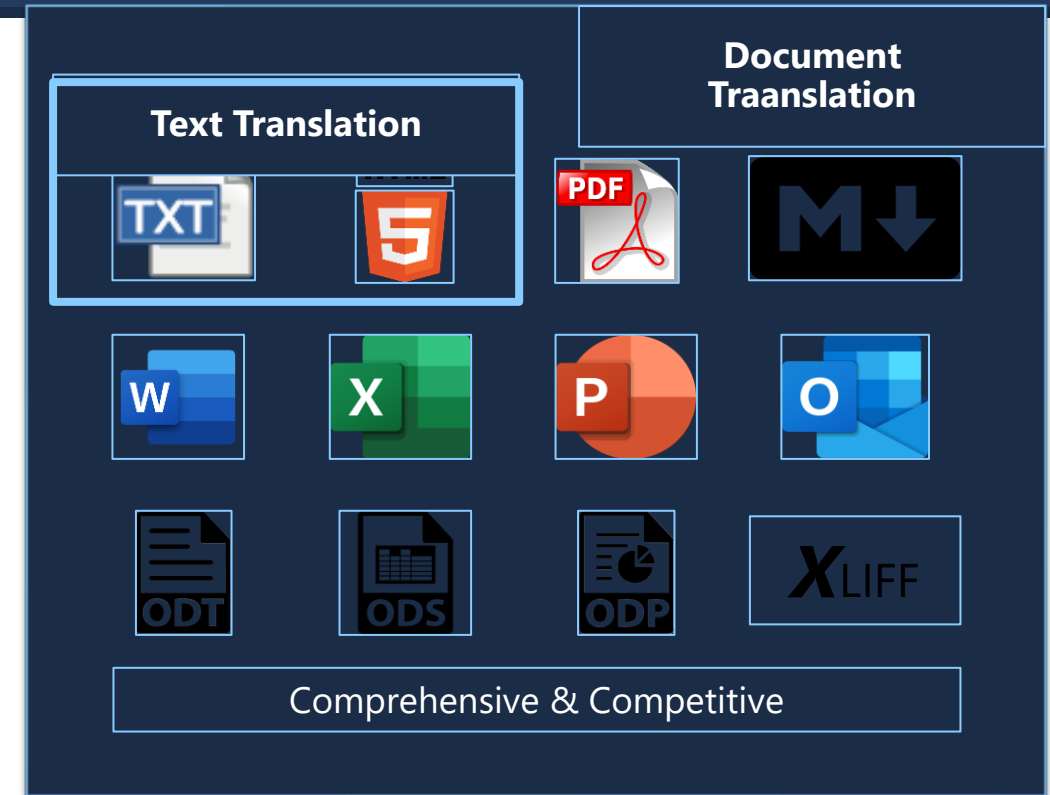
Afrikaans	Dari	Icelandic	Lingala	Punjabi	Tatar
Albanian	Divehi	Igbo	Lithuanian	Queretaro Otomi	Telugu
Amharic	Dogri NEW!	Indonesian	Luganda	Romanian	Thai
Arabic	Dutch	Inuinnaqtun	Macedonian	Rundi	Tibetan
Armenian	English	Inuktitut	Maithili	Russian	Tigrinya
Assamese	Estonian	Inuktitut (Latin)	Malagasy	Samoan	Tongan
Azerbaijani	Faroese	Irish	Malay	Serbian (Cyrillic)	Turkish
Bangla	Fijian	Italian	Malayalam	Serbian (Latin)	Turkmen
Bashkir	Filipino	Japanese	Maltese	Sesotho	Ukrainian
Basque	Finnish	Kannada	Māori	Sesotho sa Leboa	Urdu
Bhojpuri NEW!	French	Kashmiri NEW!	Marathi	Setswana	Uyghur
Bosnian (Latin)	French (Canada)	Kazakh	Mongolian (Cyrillic)	Sindhi	Uzbek (Latin)
Bodo NEW!	Galician	Khmer	Mongolian (Traditional)	Sinhala	Vietnamese
Bulgarian	Georgian	Kinyarwanda	Myanmar	Slovak	Welsh
Cantonese (Traditional)	German	Klingon	Nepali	Slovenian	Xhosa
Catalan	Greek	Klingon (plqaD)	Norwegian	Somali	Yoruba
Chinese (Literary)	Gujarati	Konkani	Nyanja	Sorbian (Lower)	Yucatec Maya
Chinese Simplified	Haitian Creole	Korean	Odia	Sorbian (Upper)	Zulu
Chinese Traditional	Hausa	Kurdish (Central)	Pashto	Spanish	
chiShona	Hebrew	Kurdish (Northern)	Persian	Swahili	
Croatian	Hindi	Kyrgyz	Polish	Swedish	
Czech	Hmong Daw	Lao	Portuguese (Brazil)	Tahitian	
Danish	Hungarian	Latvian	Portuguese (Portugal)	Tamil	

Progress in Machine Translation (MT)



Translator Overview

- Translate Text
- Translate Documents
- Translate scanned PDF documents (Image)
- Translate Speech
- Customization
 - Dictionary & Model
- Security & Compliance
- Cloud & On-premises container offering



Regulatory
Compliance



CSA



FedRAMP



GDPR



HIPAA



HITRUST



ISO



PCI



SOC



US DoD

Where is my request processed?

Region in which resource is created		Service endpoint	Request processed by data center
Translator (text)	Any	Global (recommended): api.cognitive.microsofttranslator.com	Closest available datacenter
		Americas: api-nam.cognitive.microsofttranslator.com	East US, South Central US, West Central US, and West US 2
		Europe: api-eur.cognitive.microsofttranslator.com	North Europe and West Europe
		Asia Pacific: api-apc.cognitive.microsofttranslator.com	Korea South, Japan East, Southeast Asia, and Australia East
Translator (document)	Global and any region in Americas	Custom: <name_of_your_resource>.cognitiveservices.azure.com/translator/text/batch/v1.0	East US and West US 2
	Any region in Europe		North Europe and West Europe
	Any region in Asia Pacific		Southeast Asia and Australia East

Document Translation overview

Translates rich documents retaining structure and layout

Translates documents into multiple languages

Translates batch of large documents

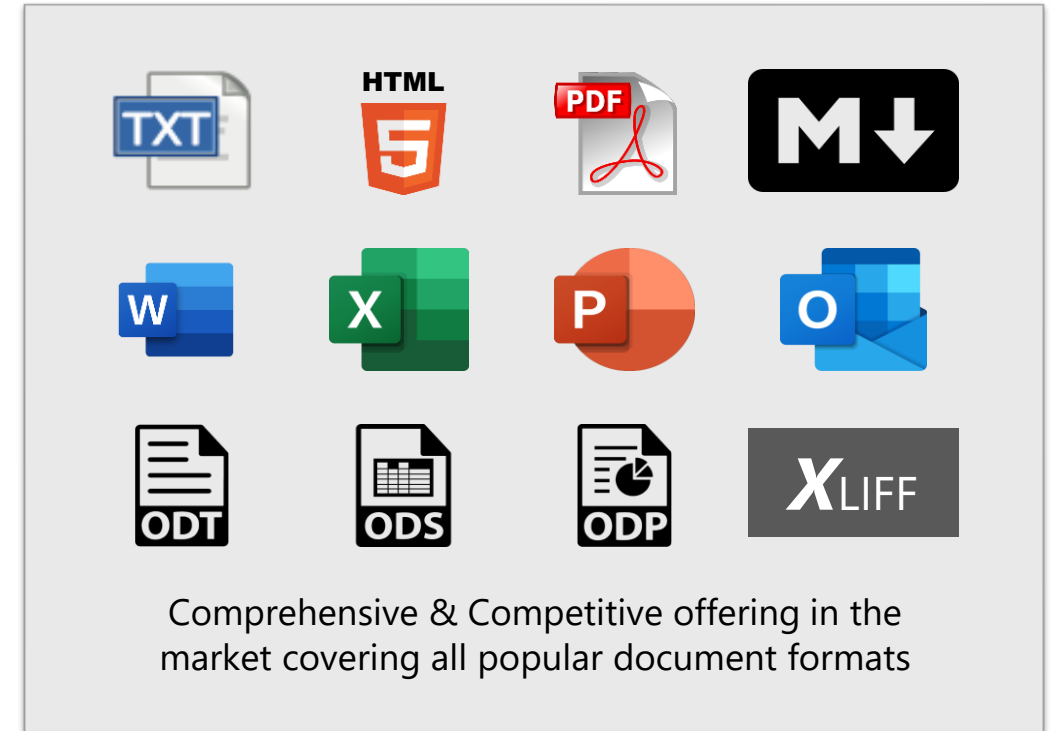
Translates multi-lingual documents

Uses general & custom models

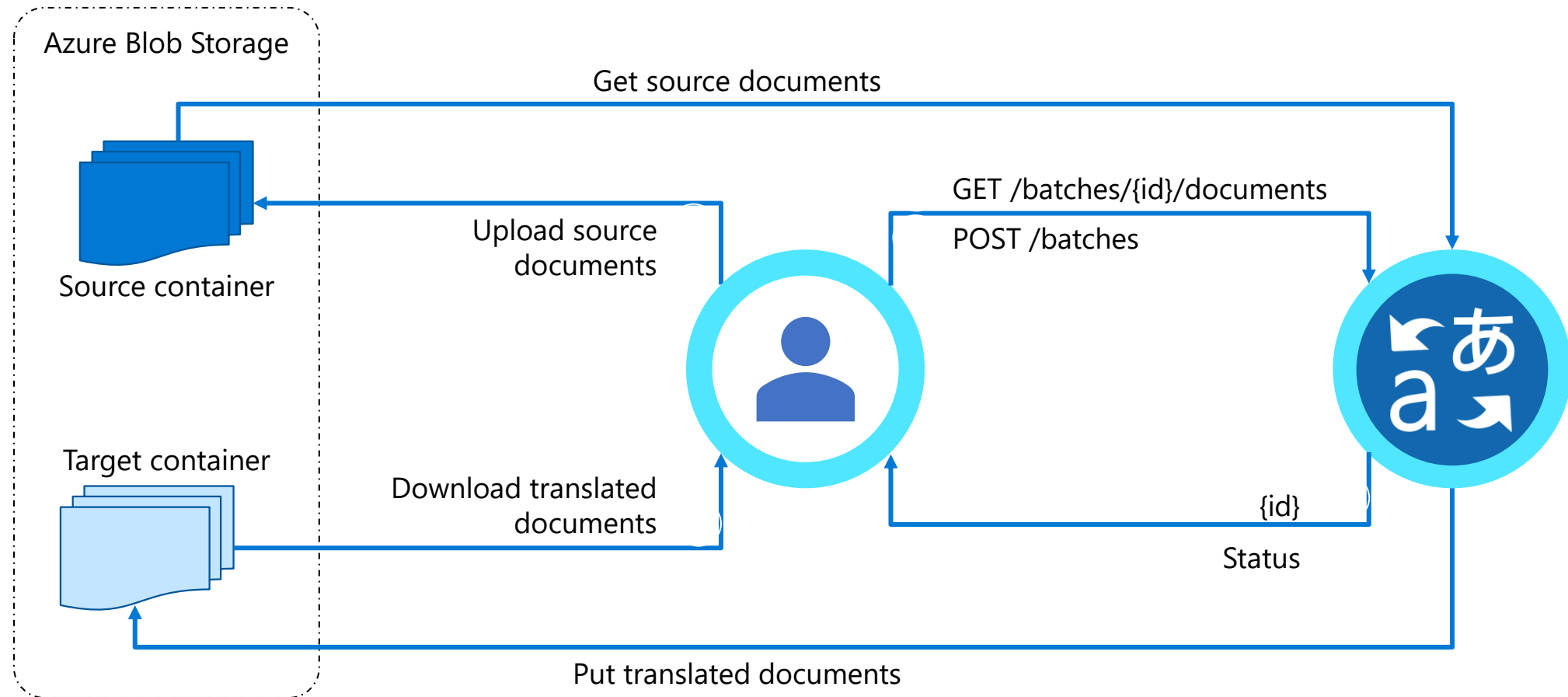
Applies custom glossaries

Azure AD authentication

Secured access to storage using managed identity



Document Translation simple flow



Document Translation

Translate at scale and preserve format

Source document

Microsoft Azure

Document Translation

Overview

Document Translation is a new feature in [Azure Translator service](#) which enables enterprises, translation agencies, and consumers who require volumes of complex documents to be translated into one or more languages preserving structure and format in the original document. It asynchronously translates whole documents in a variety of file formats including Text, HTML, Word, Excel, PowerPoint, Outlook Message, PDF, etc. across any of the 80+ languages, supported by Translator service.

Standard translation offerings in the market accept only plain text or html, and limits count of characters in a request. Users translating large documents must parse the documents to extract text, split them into smaller sections and translate them separately. **If sentences are split in an unnatural breakpoint it can lose the context resulting in suboptimal translations.** Upon receipt of the translation results, the customer must merge the translated pieces into the translated document. This involves keeping track of which translated piece corresponds to the equivalent section in the original document. The problem gets complicated when

customers want to translate complex documents having rich content.

Document Translation makes it easy for the customer to translate:

- a) volumes of large documents,
- b) documents in variety of file formats,
- c) documents requiring preserving the original layout and format, and
- d) documents into multiple target languages.

User experience

User makes a request to the Document Translation specifying location of source and target documents, and the list of target languages. It returns an identifier enabling the user to track the status of the translation. Asynchronously, Document Translation pulls each document from the source location, recognizes the document format, applies right parsing technique to extract textual content in the document, translates the textual content into target languages. It then reconstructs the translated document preserving layout and format as present in the source documents, and stores translated document in a specified location. Document Translation updates the status of translation at the document level.

Document Translation enables users to customize translation of documents by providing custom glossaries, a custom model id built using [customer translator](#), or both as part of the request. Such customization retains specific terminologies and provides domain specific translations in the translated documents.

Feature ¹	Microsoft Translator
Supported file formats	txt, tab, tsx, html, docx, pptx, xlsx, msg, pdf, tmx, and xlf
Total number of files	≤ 1000
Document size	≤ 40MB
Total content size in a batch	≤ 250 MB
# of target languages in a batch	≤ 10
# of languages supported	80+
Customer translation models	Supported
Custom glossary	Supported file formats – tsx, xlf
Size of glossary file	≤ 10 MB
# of glossary file per document	1 per language pair

¹ The limits in this table would be revised based on user feedback during public preview.

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Page 1 of 1

Translated document

Microsoft Azure

Traducción de documentos

Visión general

La traducción de documentos es una nueva característica del [servicio Azure Translator que permite a las](#) empresas, agencias de traducción y consumidores que requieren que los volúmenes de documentos complejos se traduzcan a uno o varios idiomas que preservan la estructura y el formato en el documento original. [Traduce asincrónicamente](#) documentos enteros en una variedad de formatos de archivo, incluyendo Texto, HTML, Word, Excel, PowerPoint, Mensaje de Outlook, PDF, etc. en cualquiera de los más de 80 idiomas compatibles con el servicio Translator.

Las ofertas de traducción estándar en el mercado solo aceptan texto sin formato o html, y limita el recuento de caracteres en una solicitud. Los usuarios que traducen documentos grandes deben analizar los documentos para extraer texto, dividirlos en secciones más pequeñas y traducirlos por separado. **Si las oraciones se dividen en un punto de interrupción antinatural, puede perder el contexto dando lugar a traducciones subóptimas.** Al recibir los resultados de la traducción, el cliente debe combinar las piezas traducidas en el documento traducido. Esto implica realizar un seguimiento de qué pieza traducida corresponde a la sección equivalente del documento original. El problema

se complica cuando los clientes quieren traducir documentos complejos que tienen contenido enriquecido.

La traducción de documentos facilita al cliente la traducción:

- a) volúmenes de documentos grandes,
- b) documentos en variedad de formatos de archivo,
- c) documentos que requieren preservar el diseño y el formato originales, y
- d) documentos en varios idiomas de destino.

Experiencia de usuario

User realiza una solicitud a la Traducción de documentos especificando la ubicación de los documentos de origen y de destino, y la lista de idiomas de destino. [Devuelve un identificador](#) que permite al usuario realizar un seguimiento del estado de la traducción. Asincrónicamente, Traducción de documentos extrae cada documento de la ubicación de origen, reconoce el formato del documento, aplica la técnica de análisis correcta para extraer contenido textual en el documento, traduce el contenido textual a idiomas de destino. A continuación, reconstruye el documento traducido conservando el diseño y el formato tal como está presente en los documentos de origen y almacena el documento traducido en una ubicación especificada. Traducción de documentos actualiza el estado de la traducción en el nivel de documento.

La traducción de documentos permite a los usuarios personalizar la traducción de documentos proporcionando glosarios personalizados, un identificador de modelo personalizado creado mediante [el traductor de clientes](#) ambos como parte de la solicitud. Dicha personalización conserva terminologías específicas y proporciona traducciones específicas de dominio en los documentos traducidos.

Característica ¹	Traductor de Microsoft
Formatos de archivo compatibles	txt, tabulador, tsx, html, docx, pptx, xlsx, msg, pdf, tmx y xlf
Número total de archivos	≤ 1000
Tamaño del documento	≤ 40MB
Tamaño total del contenido en un lote	≤ 250 MB
# de idiomas de destino en un lote	≤ 10
# de idiomas soportados	Más de 80
Cliente translation models	Apoyado
Glosario personalizado	Formato de archivo compatibles – tsx, xlf
Tamaño del archivo glosario	≤ 10 MB
# de archivo de glosario por documento	1 por par de idiomas

¹ Los límites de esta tabla se revisarían en función de los comentarios de los usuarios durante la vista previa pública.

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Página 1 de 1

Scanned PDF translation

Document translation now can translate PDF documents having scanned content

Document translation service internally calls OCR, extracts text, translates and reconstructs a digital PDF document

- Font formatting like bold, italics, underline, highlights, etc. aren't retained
- Currently content written right-to-left and top-to-bottom are not supported

Support of scanned PDF translation comes at no additional cost to the customer.



Translation Example



Lake Washington starts at home.

Rain water and wash water carry pollution into our creeks, Lake Washington, and Puget Sound.

Join your neighbors in taking the

KIRKLAND CLEAN WATER PLEDGE

to help protect Kirkland's creeks, lakes, and wetlands.

Sign the pledge at kirklandwa.gov/cleanwater and receive a **FREE CHINOOK BOOK** mobile coupon app subscription.

ONLY RAIN DOWN THE DRAIN

CITY OF KIRKLAND WASHINGTON



El lago Washington comienza en casa. El agua de lluvia y el agua de lavado llevan la contaminación a nuestros arroyos, el lago Washington y Puget Sound.

Únase a sus vecinos para tomar el

COMPROMISO DE AGUA LIMPIA DE KIRKLAND

para ayudar a proteger los arroyos, lagos y humedales de Kirkland.

Firme el compromiso en kirklandwa.gov/cleanwater y reciba una suscripción GRATUITA a la aplicación de cupones móviles CHINOOK BOOK.

SOLO LLUVIA POR EL DESAGÜE

CITY OF KIRKLAND WASHINGTON

Translation Example

Thanks for helping protect our waterways!



Yard Care and Maintenance

Minimize use of fertilizers and chemicals in your yard. Follow directions on label and never use more than suggested.

Do not sweep, blow, or dump yard debris into streets, sidewalks, ditches or drains. Dispose of yard waste in yard waste bins.

Did you know? More than 35 different pesticides have been detected in our local creeks. Adding just a 3-inch layer of mulch in your garden can prevent weeds as effectively as yard chemicals.



Pressure Washing

Use a broom to sweep up dirt and debris. Divert wash water to a landscaped area where wash water can soak into the ground. If you can't divert wash water, use only cold water with no soap or chemicals.

Don't allow soap, chemicals, or dirt into street or storm drain.



Roof Cleaning

Use a broom, stiff brush, or leaf blower to remove moss. If you must use chemical cleaners, disconnect roof downspouts so chemicals do not flow into the stormwater drainage system.

Direct wash water to adjacent lawn and landscape to soak into the ground.



Car Washing

Use a commercial car wash or wash your car on a lawn or gravel area. Keep soap (even biodegradable soap) and dirty water out of streets and storm drains.

Did you know? The average driveway car wash wastes more than 120 gallons of water. The average commercial car wash uses 60% less water and keeps soap and chemicals out of our streams.



Paint and Chemical Disposal

Never dump chemicals, paint, or rinse water in your yard or down storm drains. Always rinse paint brushes and rollers in a sink, never outside.

Visit hazwastehelp.org for info on safe disposal of household chemicals or paintcare.org/WA for FREE paint disposal.

If you see soap, oil, or other pollution in your neighborhood storm drains or creeks, call our 24/7 Spill Hotline:

425-587-3900

Alternative language formats of this publication are available upon request. For more information contact (425) 587-3831 or TitleViceCoordinator@kirklandwa.gov.

Learn more at kirklandwa.gov/stormwater

City of Kirkland Public Works Storm & Surface Water Division

¡Gracias por ayudar a proteger nuestras vías fluviales!



Cuidado y mantenimiento del patio

Minimice el uso de fertilizantes y productos químicos en su jardín. Siga las instrucciones en la etiqueta y nunca use más de lo sugerido. No barrer,

soplar o tirar escombros del patio en calles, aceras, zanjas o desagües. Deseche los residuos de patio en los contenedores de basura del patio.

¿Sabías que? Se han detectado más de 35 pesticidas diferentes en nuestros arroyos locales. Agregar solo un 3 pulgadas de mantillo en su jardín puede prevenir que estas sustancias tan efectivamente como los productos químicos del jardín.



Lavado a presión

Use una escoba para barrer la suciedad y los escombros. Desvíe el agua de lavado a un área ajardinada donde el agua de lavado puede sumergirse en el suelo. Si no puede desviar el agua de lavado, use solo agua fría sin jabón ni productos químicos.

No permita que el jabón, los productos químicos o la suciedad entren en la calle o en el desagüe pluvial.



Limpeza del techo

Use una escoba, un cepillo rígido o un soplador de hojas para eliminar el musgo. Si debe usar limpiadores químicos, desconecte las bajantes del techo para que los productos químicos no fluyan hacia el sistema de drenaje de aguas pluviales.

Lave directamente el agua al césped y al paisaje adyacentes para sumergirse en el suelo.



COCHE ERA

Lavado de coches

Use un lavado de autos comercial o lave su automóvil en un área de césped o grava. Mantenga el jabón (incluso el jabón biodegradable) y el agua sucia fuera de las calles y los desagües pluviales.

¿Sabías que? El lavado de autos promedio en el centro de entrada desperdicia más de 120 galones de agua. El lavado de autos comercial promedio usa un 60% menos de agua y mantiene el jabón y los productos químicos fuera de nuestros arroyos.



Pintura y eliminación de productos químicos

Nunca arroje productos químicos, pinte o enjuague el agua en su patio o en los desagües pluviales. Siempre enjuague los pinceles y rodillos en un fregadero, nunca afuera.

Visite hazwastehelp.org para obtener información sobre la eliminación segura de productos químicos domésticos o paintcare.org/WA para la eliminación GRATUITA de pintura.

Si ve jabón, aceite u otra contaminación en los desagües pluviales o arroyos de su vecindario, llame a nuestra línea directa de derrames las 24 horas del día, los 7 días de la semana:

425-587-3900

Los formatos de idiomas alternativos de esta publicación están disponibles bajo petición. Para obtener más información, comuníquese con (425) 587-3831 o TitleViceCoordinator@kirklandwa.gov.

Más información en kirklandwa.gov/stormwater

División de Tormentas y Aguas Superficiales de Obras Públicas de la Ciudad de Kirkland

Translator API limits



Max request size: 10K characters

Character limits per hour:



Tier	Character limit
F0	2 million characters per hour
S1	40 million characters per hour
S2 / C2	40 million characters per hour
S3 / C3	120 million characters per hour
S4 / C4	200 million characters per hour

Custom translation limited to 1,800 characters/second



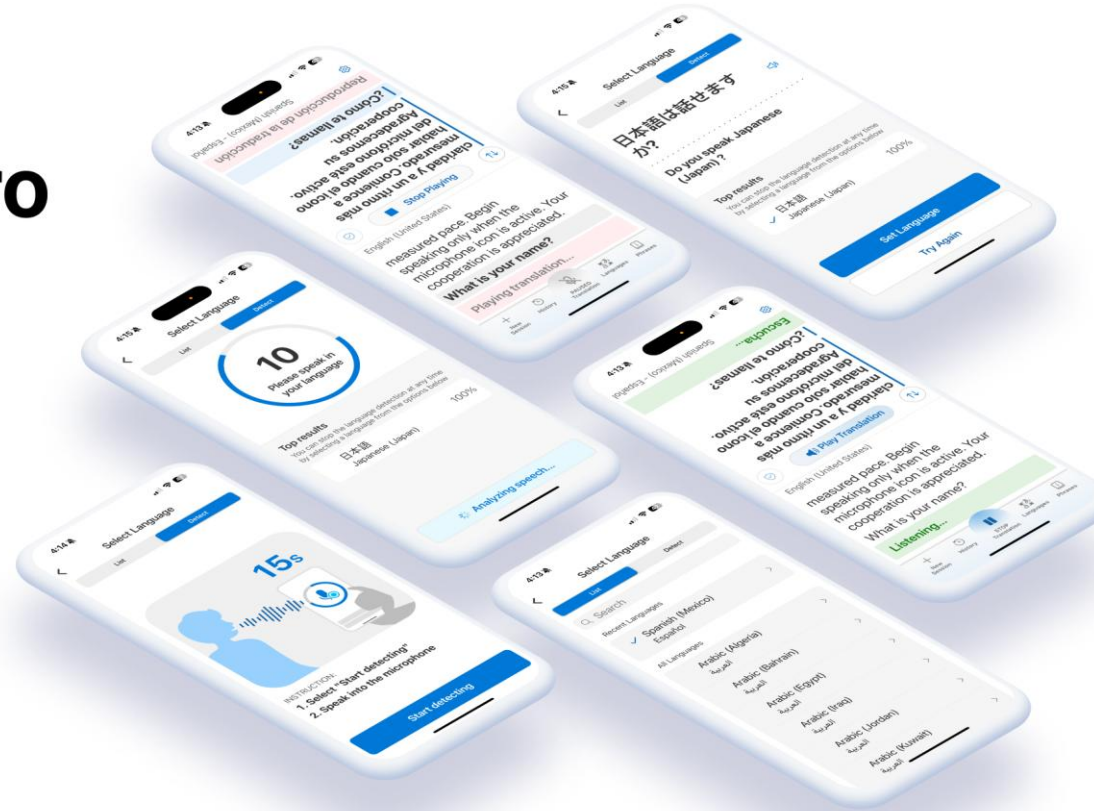
Average latency: 150 milliseconds to 300 milliseconds

Max latency: 15 seconds for standard models, 120 seconds for custom models

Microsoft Translator Pro



Microsoft Translator Pro



AI Translator Demonstration

AI Translator

The screenshot displays the Azure AI Foundry interface for the 'Document translation' service. The top navigation bar includes 'Azure AI Foundry', 'project1', 'AI Services', 'Language', and 'Document translation'. The left sidebar lists various services and tools, with 'AI Services' currently selected. The main content area is titled 'Document translation' and provides instructions on translating documents. It includes a 'Try it out' section with a dropdown for 'Connected Azure AI Services' (set to 'ai-azureaihub586273718236 (australiaeast)'), fields for 'Select source language' (set to 'Auto detect') and 'Select target language', and a large area for uploading a document or choosing from samples. The 'On this page' sidebar on the right lists 'Introduction', 'Try it out', and 'Run the code'. The 'Source document' label is at the bottom left of the main content area.

Document translation

Translate documents from source language to target language from file types such as .docx, .pptx, .xlsx, .txt, .html and more.

[View code](#)

Try it out

Connected Azure AI Services *

ai-azureaihub586273718236 (australiaeast)

[Create a new AI Services resource](#)

Select source language * Select target language *

Auto detect

Upload document or choose from the samples *

Drag and drop file here or [Browse for a file](#)

Word
Overview of document translation feature in Azure AI

Power Point
Azure AI Translator's features for enterprise customers

Text file
Overview of medical report of an accident

Text file
Travel blog of a couple in Los Angeles

Source document

[Management center](#)

On this page

- Introduction
- Try it out
- Run the code

Which language service should I use?

Which Language service feature should I use?

What do you want to do?	Document format	Recommended Solution	Is this solution customizable?*
Detect and/or redact sensitive information such as PII and PHI.	Unstructured text, transcribed conversations	PII detection	
Extract categories of information without creating a custom model.	Unstructured text	The preconfigured NER feature	
Extract categories of information using a model specific to your data.	Unstructured text	Custom NER	✓
Extract main topics and important phrases.	Unstructured text	Key phrase extraction	
Determine the sentiment and opinions expressed in text.	Unstructured text	Sentiment analysis and opinion mining	
Summarize long chunks of text or conversations.	Unstructured text, transcribed conversations.	Summarization	
Disambiguate entities and get links to Wikipedia.	Unstructured text	Entity linking	
Classify documents into one or more categories.	Unstructured text	Custom text classification	✓
Extract medical information from clinical/medical documents, without building a model.	Unstructured text	Text analytics for health	
Build a conversational application that responds to user inputs.	Unstructured user inputs	Question answering	✓
Detect the language that a text was written in.	Unstructured text	Language detection	
Predict the intention of user inputs and extract information from them.	Unstructured user inputs	Conversational language understanding	✓
Connect apps from conversational language understanding, LUIS, and question answering.	Unstructured user inputs	Orchestration workflow	✓

Should I choose Azure AI Language or Azure Open AI?

AI Language

- Targeted focus and capabilities
- Deterministic and repeatable
- Available as a managed service or containerized (in some cases) to run locally/securely
- Excels in scale and low latency
- Customizable
- Returns a confidence score as part of the response
- No prompting required

Azure OpenAI

- Broad capabilities focused on generation and natural language interactions
- Not inherently deterministic
- Hosting/Fine tuning requires large amounts of compute so generally used as a managed service
- Highly capable in language tasks

Responsible Use

Responsible AI Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of Responsible AI
- ✓ How to access transparency notes for AI Language services

2

Identify....

- ✓ Monitoring and Logging options for your AI Language tools



Microsoft's Responsible AI principles



Fairness



Reliability & Safety



Privacy & Security



Inclusiveness



Transparency



Accountability

Learn more at aka.ms/RAI

Responsible AI / Transparency Notes

- [Transparency Note for Azure AI Language](#)
- [Transparency note for Named Entity Recognition and Personally Identifying Information](#)
- [Transparency note for the health feature](#)
- [Transparency note for Key Phrase Extraction](#)
- [Transparency note for Language Detection](#)
- [Transparency note for Question answering](#)
- [Transparency note for Summarization](#)
- [Transparency note for Sentiment Analysis](#)
- [Azure AI Translator Transparency Note](#)

Data, Privacy and Security

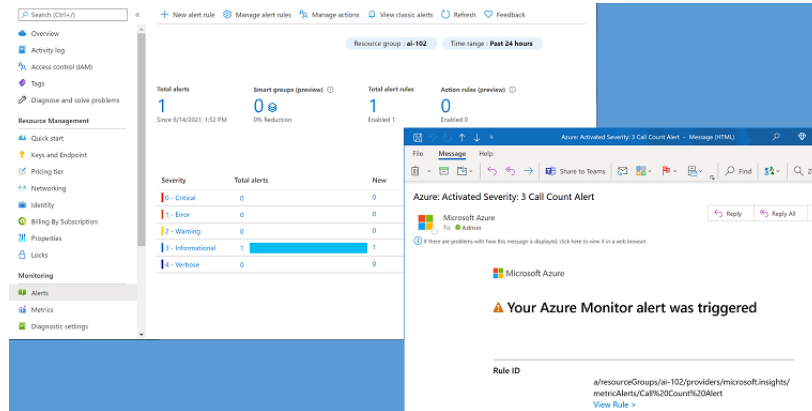
What data does Azure AI Language process and how does it process it?

- Azure AI Language processes text data that is sent by the customer
- All results are sent back to the customer in the API response
- Azure AI Language uses aggregate telemetry
- Azure AI Language doesn't store or process customer data outside the region
- Azure AI Language encrypts all content, including customer data, at rest.

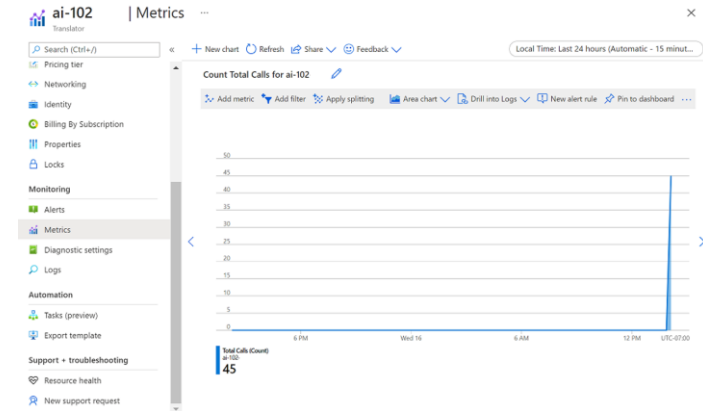
How is data retained and what customer controls are available?

- Data sent in synchronous or asynchronous calls may be temporarily stored by Azure AI Language for up to 48 hours only and is purged thereafter.
- To prevent this temporary storage of input data, the LoggingOptOut query parameter can be set accordingly.

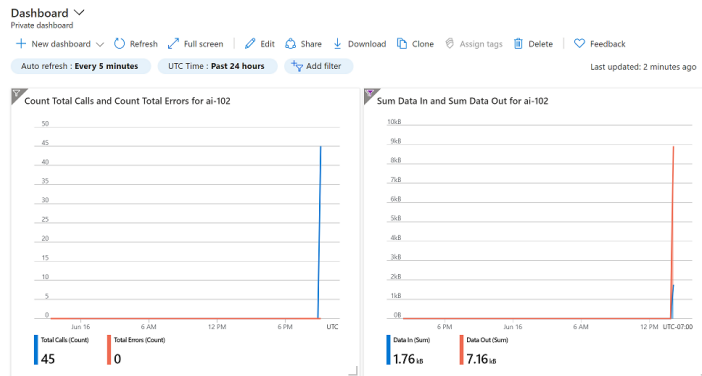
Monitoring and Logging



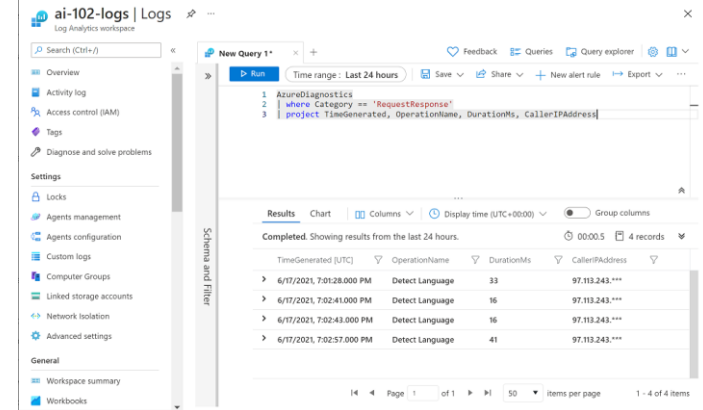
Create Cost Alerts



View Metrics



Create a Dashboard



Manage Diagnostic Logging

Containerization, Integration, Examples and Task Agents

Containerization Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ The principles of containerized language services
- ✓ Sample architectures and use cases for the AI Language service

2

Identify....

- ✓ Where Task Agents could make use of the AI Language service



Containerized language services



Azure AI task-specific models and capabilities

Speech



Enable every device to hear and speak

- Speech to text
- Text to speech
- Custom Neural Voice
- Speech Translation
- Speaker Diarization
- Language Identification

Translator



Understand any language

- Text translation
- Document translation
- Microsoft Translator Pro (preview)
- Embedded translation

Language



Extract information & process intents

- Summarization
- Entity extraction (PII/PHI)
- Conversational AI
- Text analytics
- Sentiment mining

Content Understanding



Understand multimodal contents

- Document Intelligence
- Image analysis
- Video analysis
- Audio analysis
- Facial recognition

Responsible AI



Ensure AI use remains safe

- Content safety filters
- Safety evaluations

AI task-specific models available in



Microsoft
Public Cloud



Microsoft
Government Cloud



Connected
Containers



Disconnected
Containers

Regulatory
Compliance



HITRUST



Customer pain points

Data security & privacy concerns



Unable to upload all audio/text data to the cloud



Low bandwidth or intermittent connectivity



Azure AI Container Capabilities



Immutable infrastructure | Adapt to changes and avoid configuration drift



Control over data | Choose where data is processed with drive data, management, identity & security consistency



Control over model updates | Flexible versioning and control over when to deploy in your environments



Portable architecture | Create portable app architecture to deploy on Azure, on-prem, and edge endpoints



High throughput, Low latency | Run apps physically close to data and you control the throughput

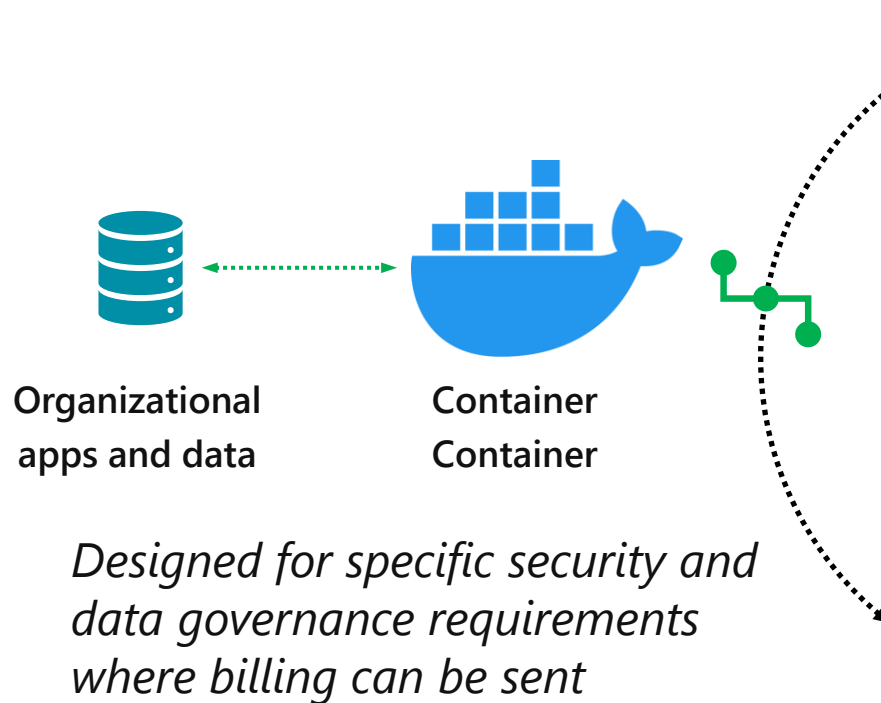


Scalability | Build your own apps on a scalable cluster foundation based on your business requirements

AI Container environment options

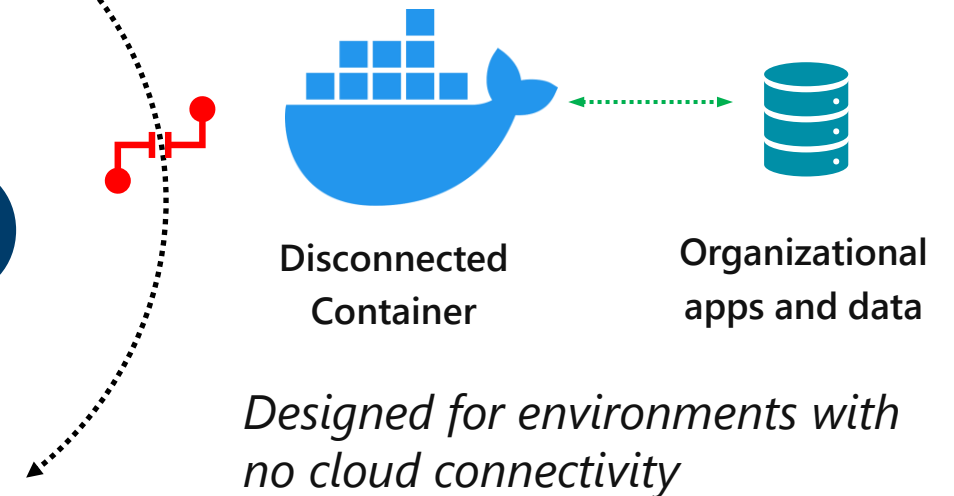
Pay-as-you-go consumption model where **only** billing data is sent to the cloud

- **Connected** containers



- **Disconnected** containers

Pay up front for a years' worth of consumption with **no** data sent to the cloud



AI task-specific models available in Containers



Speech

Speech to Text
Text to Speech
Neural Speech to Text
Custom Text to Speech*
Speech Language
Identification (Preview)*
Speaker Diarization*



Translator

Text Translation
Document Translation
Transliteration



Language

Language Detection
Key Phrase Extraction
Sentiment Analysis
PII Redaction
Summarization
CLU
NER
Custom NER*
Text Analytics for Health*



Content Understanding

Document Intelligence
includes OCR - designed for
documents

Computer vision OCR
optimized for images*



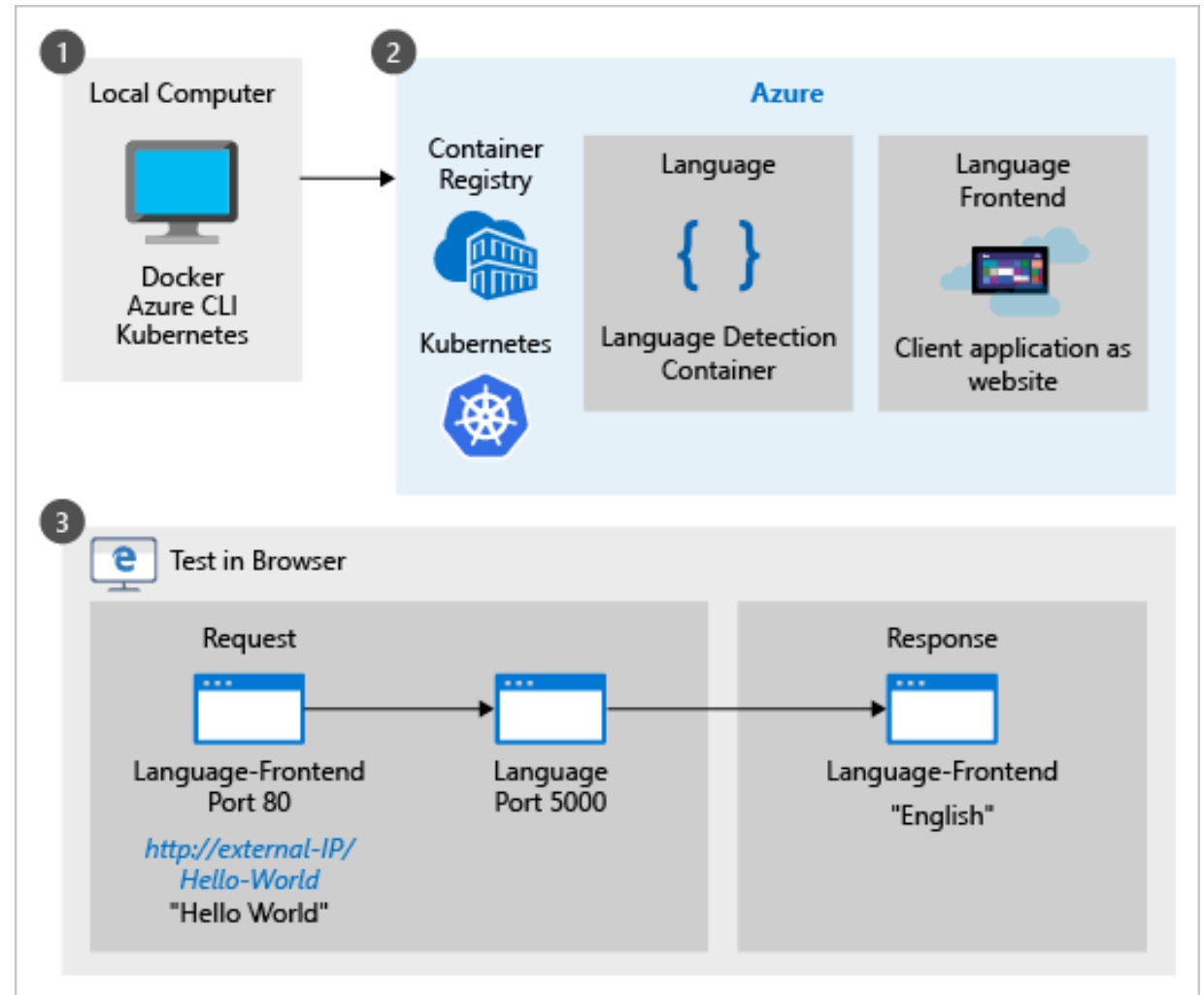
Responsible AI

Content Safety (limited access preview)

* **Available** *only* in connected containers

Deploy a language detection container to Azure Kubernetes Service

<https://learn.microsoft.com/en-us/azure/ai-services/containers/azure-kubernetes-recipe>

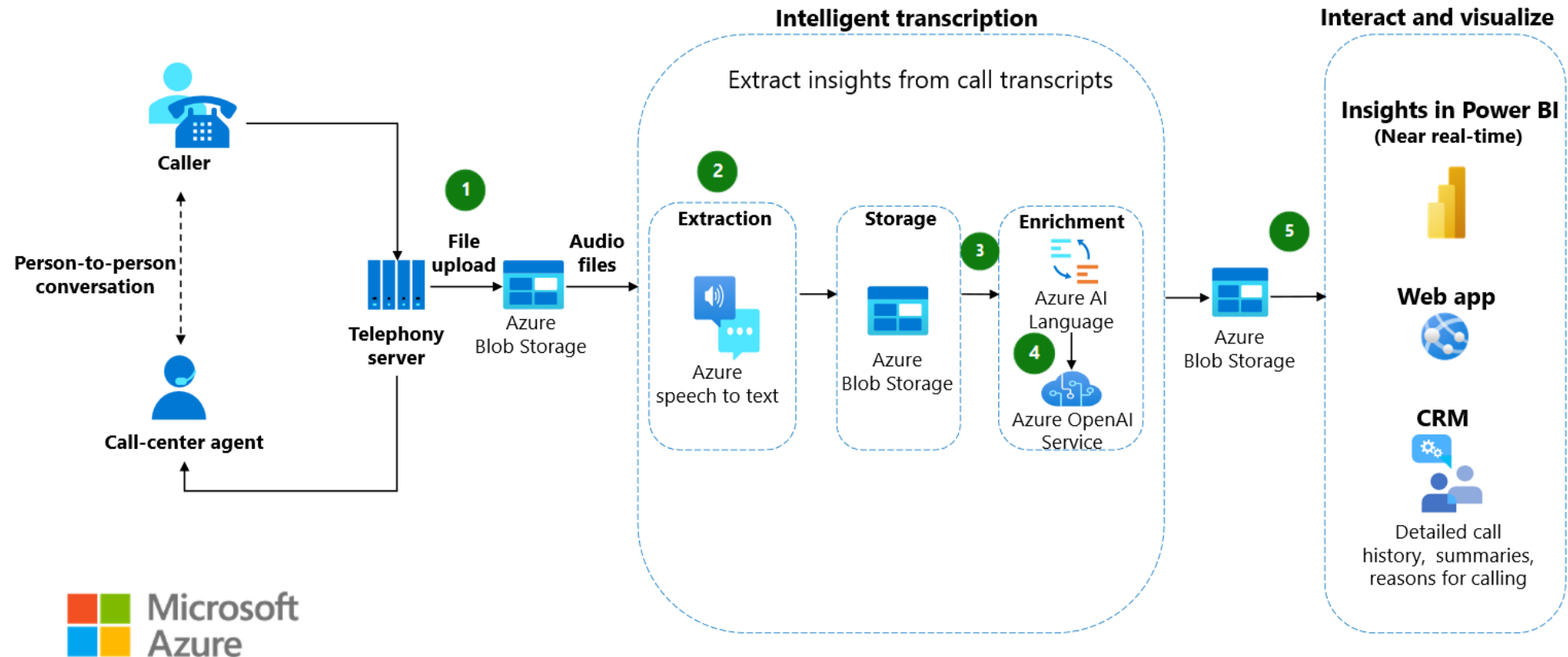


Example: Accelerator for disconnected AI Services



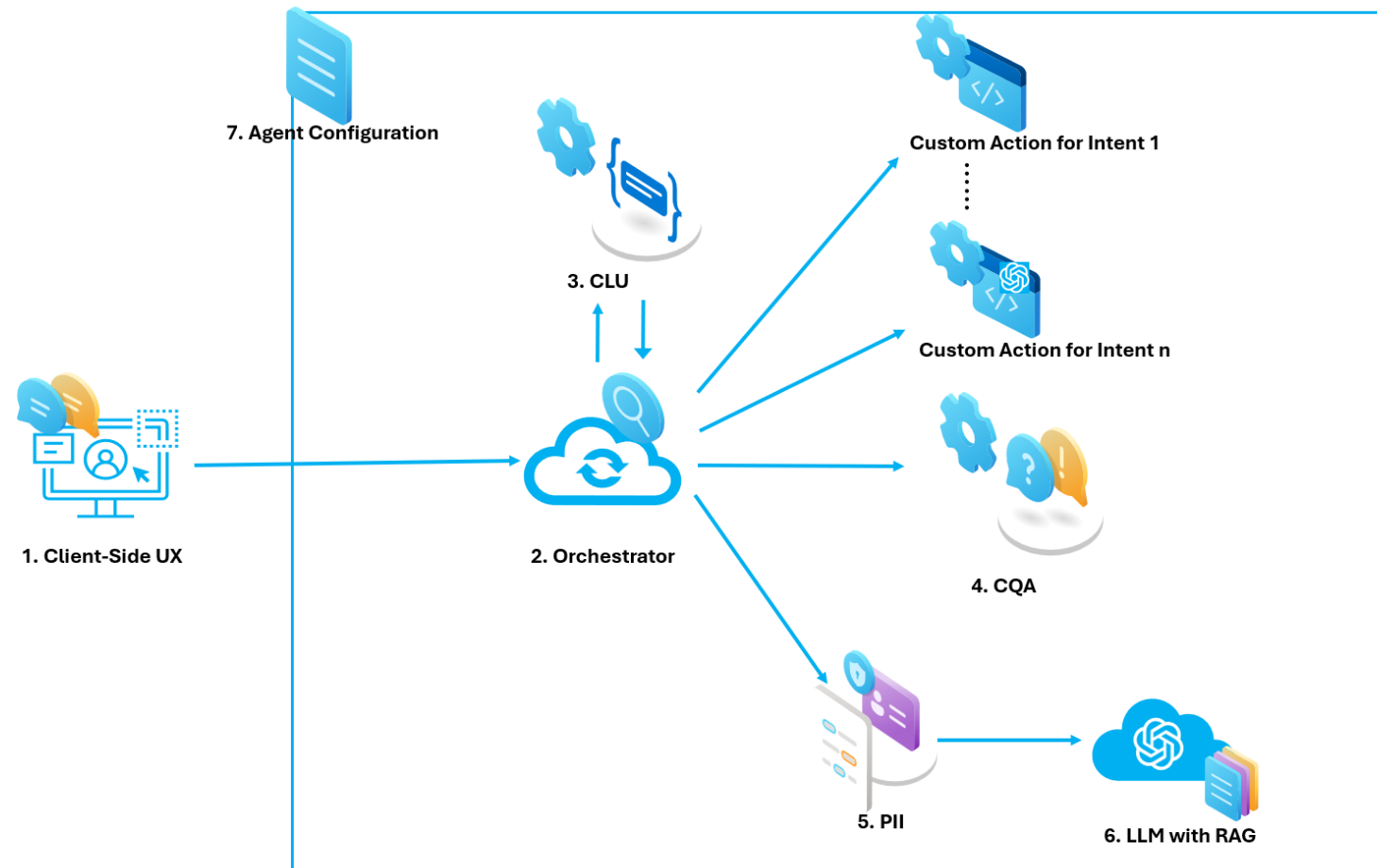
Sample Architectures / Examples

Extract and analyze call center data

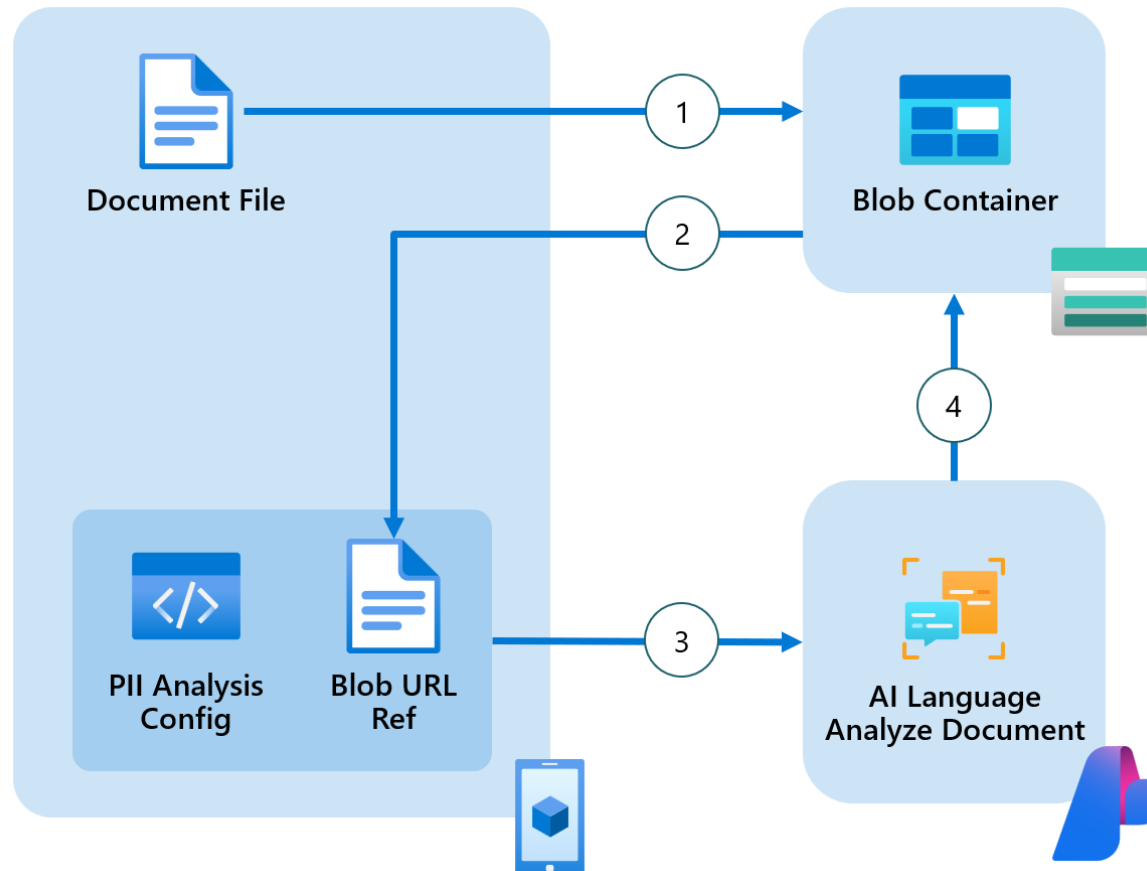


Azure Language OpenAI Conversational Agent Accelerator

Reference Architecture



Document Redaction with Azure AI Language PII Native Document Analysis



Triage incoming emails with custom text classification and power automate

The screenshot displays the Microsoft Power Automate web interface. The top navigation bar is blue with the 'Power Automate' logo, a search bar, and user information. The left sidebar contains navigation links: Home, Approvals, My flows (selected), Create, Templates, Connectors, Data, Monitor, AI Builder, Process advisor, Solutions, and Learn. The main workspace shows a flow named 'EmailTriage'. The first step is 'When a new email arrives (V3)' with the folder set to 'Inbox'. The second step is 'Async CustomSingleLabelClassification (2022-05-01)'. This step has three input fields: 'documents id - 1' with the value '1', 'documents text - 1' with the value 'Body', and 'documents language - 1' with a placeholder text. Below these is an 'Add new item' button. At the bottom, there are two more input fields: 'projectName' with the value 'WebOfScience' and 'deploymentName' with the value 'production'.

Power Automate

Search

Environments MS Personal Productiv...

Undo Redo Comments Save Flow checker Test

EmailTriage

When a new email arrives (V3)

Folder Inbox

Show advanced options

Async CustomSingleLabelClassification (2022-05-01)

*documents id - 1
1

*documents text - 1
Body

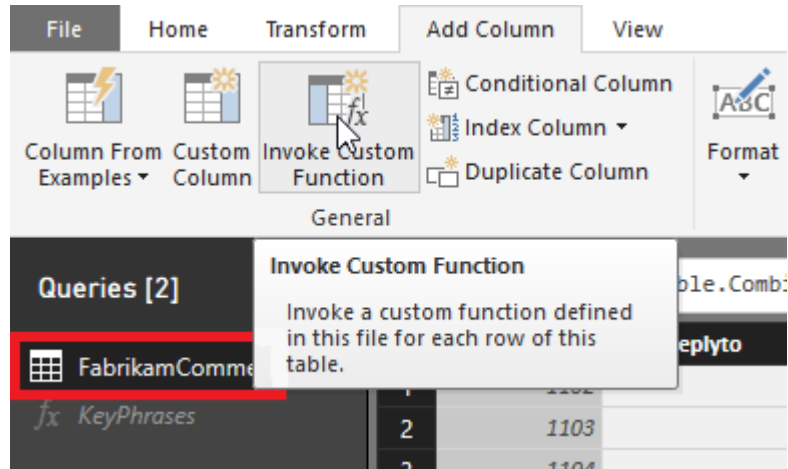
documents language - 1
(Optional) This is the 2 letter ISO 639-1 representation of a language. For example, use "en"

+ Add new item

*projectName WebOfScience

*deploymentName production

Extract key phrases from text stored in Power BI





<https://aka.ms/language-accelerators>

Beirrsdorf



Azure AI Language



Azure AI Search

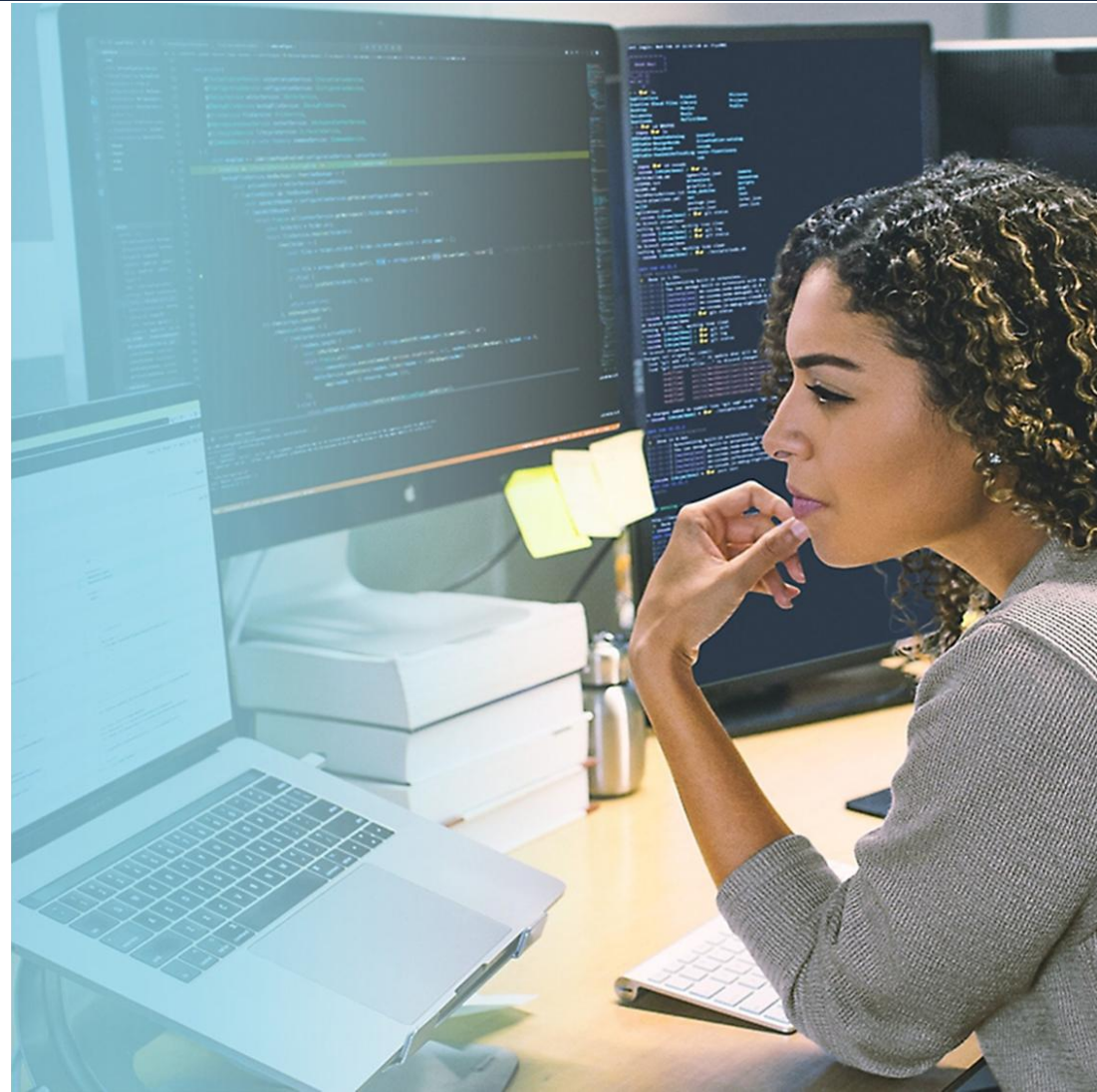


Azure AI Services

Beiersdorf

"We're able to free our researchers from interacting with dozens of different interfaces, which means more quality time spent on research and innovation."

Stephan Abend, Head of the Data Science Hub, Beiersdorf



Arthur Little



Azure AI Language



Azure OpenAI Service



Azure AI Search

ARTHUR  LITTLE

“We were surprised that Microsoft Azure offers such a breadth of advanced AI capabilities, like abstract summarization, which are way beyond even the specialist AI providers”

- Jon Nicholls, Global CIO, Arthur D. Little



Datasite

 Azure AI Language  Azure AI Translator  Azure AI Services



"Sellers might have up to 100,000 documents associated with a deal.... AI features enabled by Azure AI Language can potentially compress weeks of work into days for our customers."

Thomas Frieden, Chief Product Officer, Datasite



Task Agents

Azure AI Foundry



Copilot Studio



Visual Studio



GitHub



Azure AI
Foundry SDK



Model Catalog

Foundational models

Open-source models

Task models

Industry models



Azure
OpenAI Service



Azure
AI Search



Azure AI
Agent Service



Azure AI
Content Safety



Azure Machine
Learning

Evaluations

Customization

Governance

Monitoring

Observability

Agentic AI

A new frontier

What are agents?

AI designed to perform a task

Tasks can vary in level of complexity and capabilities depending on your need

Simple



Generation

Generate summaries, images, audio, and more with an AI model and inputs.

Generally available



Retrieval

Retrieve information from grounding data, reason, summarize, and answer user questions

Generally available



Action

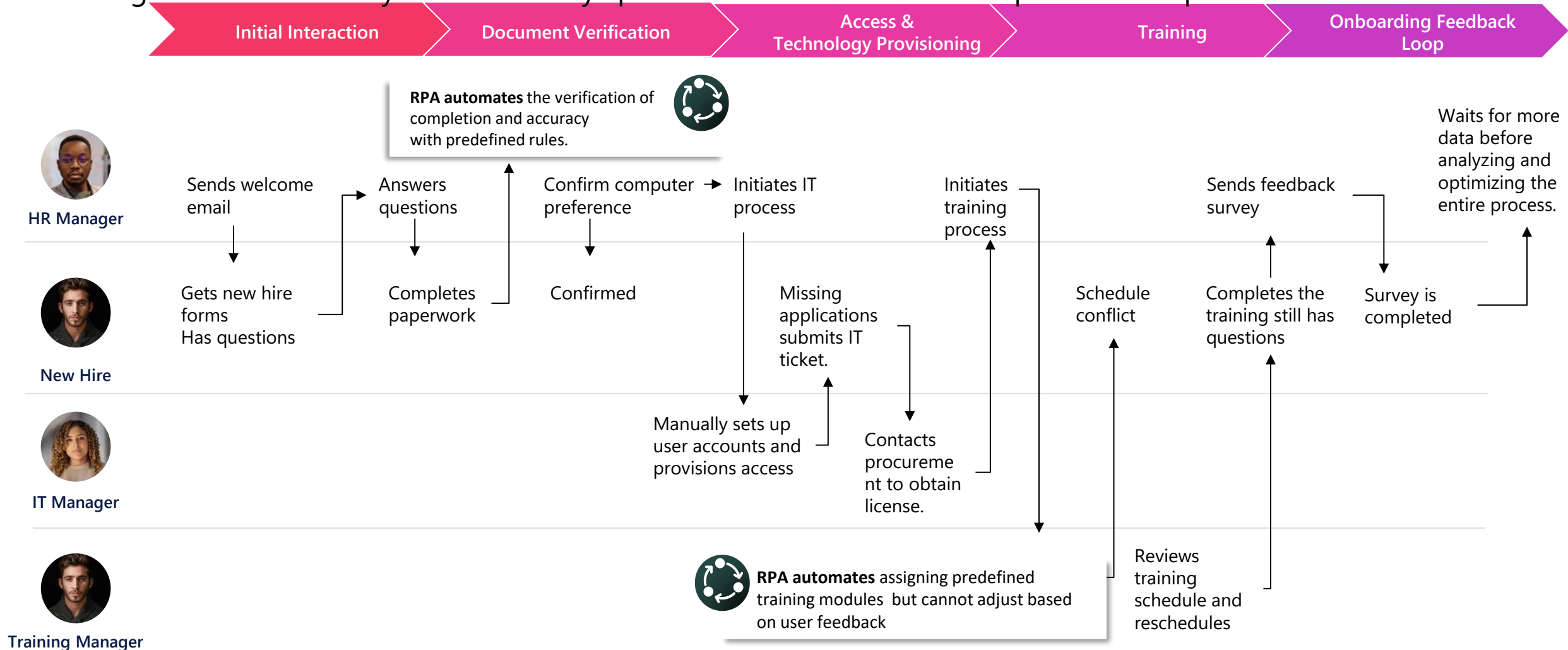
Take actions to automate workflows, and replace repetitive tasks for users

Generally available

Advanced

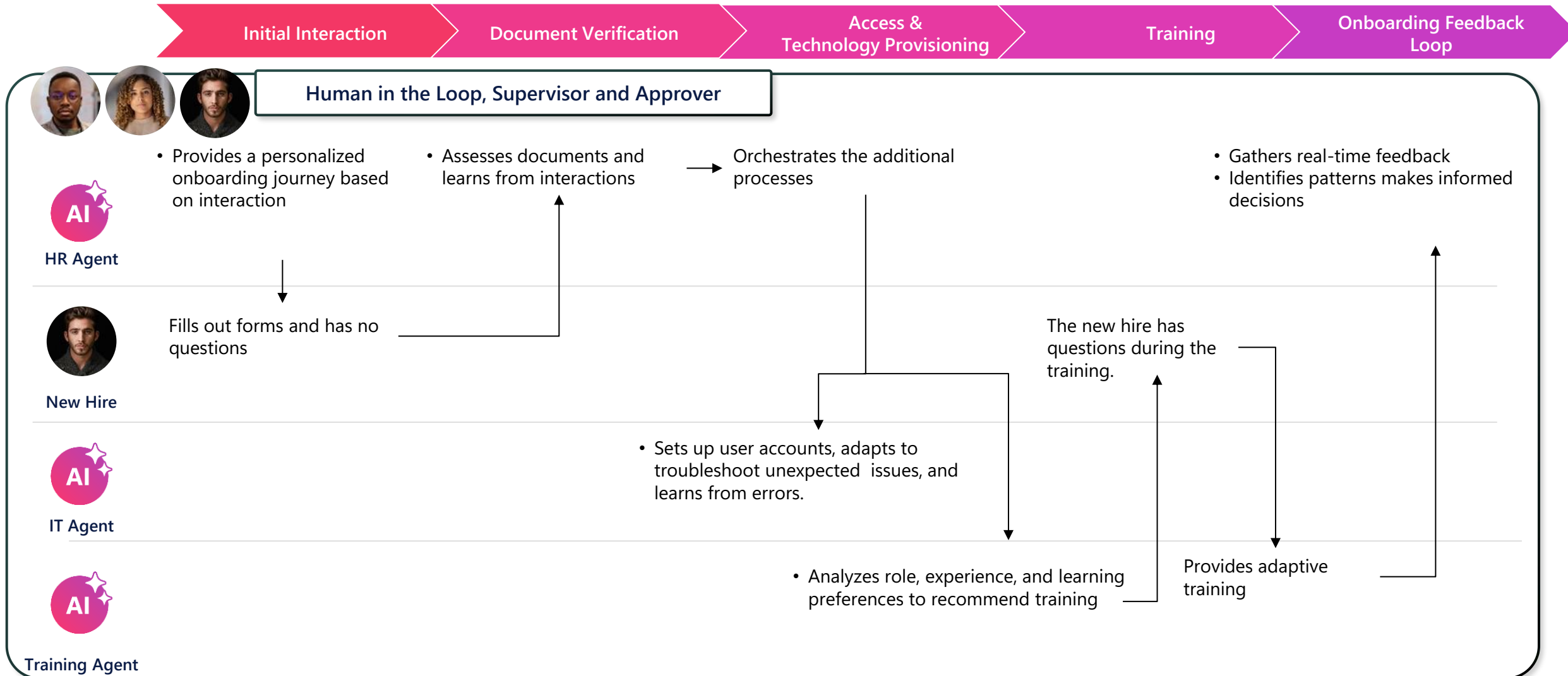
TODAY

Existing solutions can only automate very specific tasks that have clear inputs and outputs



TOMORROW

With AI Agents, these steps can be automated for the first time, while keeping human in the loop.



Interpreter Agent in Teams

The screenshot displays the Microsoft Teams interface during a meeting. The top toolbar includes icons for Chat, People, Raise, React, View, Notes, Rooms, Copilot, and a 'More' menu (three dots). A red callout '1' points to the 'More' menu. The 'More' menu is open, showing options: Add an app, Record and transcribe, Meeting info, Video effects and settings, Audio settings, Language and speech, Settings, Call me, and Help. A red callout '2' points to the 'Language and speech' option. Below the 'More' menu, a secondary menu is open, showing: Show live captions, Turn on Interpreter, and Turn on Speaker Coach. A red callout '3' points to the 'Turn on Interpreter' option. A small tooltip 'Turn on Interpreter' is visible over the 'Turn on Interpreter' option. The background shows a blurred video feed of a person and the text 'Waiting for others to join...'.

00:41

Chat People Raise React View Notes Rooms Copilot More Camera Mic Share

- + Add an app
- Record and transcribe >
- i Meeting info
- 🖼️ Video effects and settings
- 🔊 Audio settings
- 🗣️ Language and speech >
- ⚙️ Settings >
- 📞 Call me
- 🔍 Help

- 📄 Show live captions
- 🗣️ Turn on Interpreter
- 🗣️ Turn on Speaker Coach

Turn on Interpreter

Waiting for others to join...

Learn More

- Azure AI Language homepage: <https://aka.ms/azure-language>
- Azure AI Language product documentation: <https://aka.ms/language-e-docs>
- Azure AI Language product demo videos: <https://aka.ms/language-videos>
- Explore Azure AI Language in Azure AI Studio: <https://aka.ms/AzureAiLanguage>

Self-paced Training

- [Certification: AI Fundamentals](#)
- [Learning Path: Azure AI Services](#)
- [Learning Path: Natural Language Processing](#)
- [Module: Monitor Azure AI services](#)

Lab Introduction

Lab Objectives

After completing this Learning, you will be able to:

1

Understand....

- ✓ How to call the AI Language service through a low code mechanism (Logic Apps)
- ✓ How to integrate Logic Apps calls with the AI Agents service in Azure AI Foundry

